

DIFFERENTIATION



SYNOPSIS

Derivative:

- (i) A function $y = f(x)$ is said to be differentiable if $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ exists finitely. This

limit is usually denoted by $f'(x)$ or $\frac{dy}{dx}$.

- (ii) Let $y = f(x)$ be a function and if $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ exists, then f is said to be differentiable at 'a' and the limit is called the derivative of f at a . The derivative of ' f ' at a is denoted by any one of the forms

$$\left(\frac{dy}{dx} \right)_{x=a} \quad (\text{or}) \quad f'(a)$$

Sum or difference rule:

- $\frac{d}{dx}(c_1 f(x) \pm c_2 g(x)) = c_1 \frac{d}{dx}(f(x)) \pm c_2 \frac{d}{dx}(g(x))$
where c_1, c_2 are constants.

Product Rule:

- i) If $f(x), g(x)$ are two differentiable functions of 'x' then

$$\frac{d}{dx}(f(x) \cdot g(x)) = f(x) \frac{d}{dx}(g(x)) + g(x) \frac{d}{dx}(f(x))$$

- ii) If $f(x), g(x)$ and $h(x)$ are three differentiable functions of 'x' then

$$\frac{d}{dx}(f(x) \cdot g(x) \cdot h(x)) = g(x)h(x) \frac{d}{dx}(f(x)) +$$

$$f(x)h(x) \frac{d}{dx}(g(x)) + f(x)g(x) \frac{d}{dx}(h(x))$$

Quotient Rule:

- If $f(x), g(x)$ are two differentiable functions of

'x' then $\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{g(x) \frac{d}{dx}(f(x)) - f(x) \frac{d}{dx}(g(x))}{(g(x))^2}$

W.E-1: $\frac{d}{dx} \left[(x+1)(x^2+1)(x^4+1)(x^8+1) \right] =$

$$(15x^p - 16x^q + 1)(x-1)^{-2} \Rightarrow (p, q) =$$

Sol: $f(x) = (x+1)(x^2+1)(x^4+1)(x^8+1) = \frac{x^{16}-1}{x-1}$

$$\Rightarrow f'(x) = \frac{16x^{15}(x-1) - (x^{16}-1)}{(x-1)^2}$$

$$p = 16, q = 15$$

Chain Rule:

- If 'y' is a function of 't' and 't' is a function of 'x' i.e., $y = f(t)$ and $t = g(x)$ then

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

similarly, if $y = f(u)$, where $u = f(v)$, $v = h(x)$ then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}$$

Note: We can extend this rule to any number of functions.

W.E-2: $\frac{d}{dx} \left(\sin^2(\log \sqrt{x}) \right) =$

Sol: $2 \sin(\log \sqrt{x}) \cos(\log \sqrt{x}) \cdot \frac{1}{\sqrt{x}} \cdot \frac{1}{2\sqrt{x}}$

Derivatives of Basic elementary functions:

- i) $\frac{d}{dx}(\text{Constant}) = 0$

ii) $\frac{d}{dx}(x^n) = n \cdot x^{n-1}$

iii) $\frac{d}{dx}(e^x) = e^x$

iv) $\frac{d}{dx}(a^x) = a^x \cdot \log a$

v) $\frac{d}{dx}(|x|) = \frac{|x|}{x}; \text{ if } x \neq 0$

$$\text{vi)} \frac{d}{dx}(\log_e |x|) = \frac{1}{x}$$

$$\text{vii)} \frac{d}{dx}(\log_a |x|) = \frac{1}{x \log a}$$

$$\text{viii)} \frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$$

$$\text{ix)} \frac{d}{dx}\left(\frac{1}{x}\right) = \frac{-1}{x^2}$$

$$\text{x)} \frac{d}{dx}\left(\frac{1}{x^n}\right) = \frac{-n}{x^{n+1}}$$

$$\text{xi)} \frac{d}{dx}([x]) = \begin{cases} 0, & \forall x \notin \mathbb{Z} \\ \text{does not exist,} & \forall x \in \mathbb{Z} \end{cases}$$

where $[\cdot]$ stands for greatest integer function.

W.E-3: For a real number 'y', Let $[y]$ denote the integral part of 'y'. Then derivative of the function

$$f(x) = \frac{\tan[x - \pi]\pi}{1 + [x]^2} \text{ is}$$

Sol: $[(x - \pi)]\pi$ is an integral multiple of π ,

$$\text{hence } f(x) = 0 \Rightarrow f'(x) = 0$$

Derivatives of Trigonometric functions:

$$\text{➤ i)} \frac{d}{dx}(\sin x) = \cos x$$

$$\text{ii)} \frac{d}{dx}(\cos x) = -\sin x$$

$$\text{iii)} \frac{d}{dx}(\tan x) = \sec^2 x$$

$$\text{iv)} \frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$$

$$\text{v)} \frac{d}{dx}(\sec x) = \sec x \cdot \tan x$$

$$\text{vi)} \frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cdot \cot x$$

Derivatives of Inverse Trigonometric functions :

$$\text{➤ i)} \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}, x \in (-1, 1)$$

$$\text{ii)} \frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}, x \in (-1, 1)$$

$$\text{iii)} \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}, x \in \mathbb{R}$$

$$\text{iv)} \frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}, x \in \mathbb{R}$$

$$\text{v)} \frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x| \cdot \sqrt{x^2 - 1}}, x < -1 \text{ or } x > 1$$

$$\text{vi)} \frac{d}{dx}(\operatorname{Cosec}^{-1} x) = \frac{-1}{|x| \cdot \sqrt{x^2 - 1}}, x < -1 \text{ or } x > 1$$

W.E-4 : If $y = \tan^{-1} \sqrt{\frac{1-x}{1+x}}$, then $\frac{dy}{d(\cos^{-1} x)} =$

Sol: Let $x = \cos 2\theta \Rightarrow \theta = \frac{1}{2} \cos^{-1} x$;

$$(-1 < x < 1)$$

$$y = \tan^{-1} \sqrt{\frac{1 - \cos 2\theta}{1 + \cos 2\theta}}$$

$$= \tan^{-1}(\tan \theta) = \theta = \frac{1}{2} \cos^{-1} x$$

$$\therefore y = \frac{1}{2} \cos^{-1} x \Rightarrow \frac{d(y)}{d(\cos^{-1} x)} = \frac{1}{2}$$

Derivatives of Hyperbolic functions:

$$\text{➤ i)} \frac{d}{dx}(\sinh x) = \cosh x$$

$$\text{ii)} \frac{d}{dx}(\cosh x) = \sinh x$$

$$\text{iii)} \frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$$

$$\text{iv)} \frac{d}{dx}(\coth x) = -\operatorname{cosech}^2 x$$

$$\text{v)} \frac{d}{dx}(\sec hx) = -\sec hx \tanh x$$

$$\text{vi) } \frac{d}{dx}(\operatorname{cosech} x) = -\operatorname{cosech} x \cdot \coth x$$

Derivatives of Inverse Hyperbolic functions:

- i) $\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1+x^2}}$
- ii) $\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2-1}}$, for $x \notin (-1, 1)$
- iii) $\frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1-x^2}$, for $x \in (-1, 1)$
- iv) $\frac{d}{dx}(\coth^{-1} x) = \frac{1}{1-x^2}$, for $x \in (-\infty, -1) \cup (1, \infty)$
- v) $\frac{d}{dx}(\operatorname{sech}^{-1} x) = \frac{-1}{x\sqrt{1-x^2}}$, for $x \in (0, 1)$
- vi) $\frac{d}{dx}(\operatorname{cosec} h^{-1} x) = \frac{-1}{|x|\sqrt{1+x^2}}$
for $x \in (-\infty, 0) \cup (0, \infty)$

Derivative of a Determinant :

➤ If $y = \begin{vmatrix} u(x) & v(x) & w(x) \\ p(x) & q(x) & r(x) \\ \lambda(x) & \mu(x) & \gamma(x) \end{vmatrix}$ then $\frac{dy}{dx} =$

$$\begin{vmatrix} u^1(x) & v^1(x) & w^1(x) \\ p(x) & q(x) & r(x) \\ \lambda(x) & \mu(x) & \gamma(x) \end{vmatrix} + \begin{vmatrix} u(x) & v(x) & w(x) \\ p^1(x) & q^1(x) & r^1(x) \\ \lambda(x) & \mu(x) & \gamma(x) \end{vmatrix} + \begin{vmatrix} u(x) & v(x) & w(x) \\ p(x) & q(x) & r(x) \\ \lambda^1(x) & \mu^1(x) & \gamma^1(x) \end{vmatrix}$$

$$+ \begin{vmatrix} u(x) & v(x) & w(x) \\ p(x) & q(x) & r(x) \\ \lambda^1(x) & \mu^1(x) & \gamma^1(x) \end{vmatrix} \text{ or similarly column wise}$$

W.E-5.

$$f(x) = \begin{vmatrix} 2 \cos x & 1 & 0 \\ x - \frac{\pi}{2} & 2 \cos x & 1 \\ 0 & 1 & 2 \cos x \end{vmatrix} \Rightarrow f^1(\pi) =$$

[EAM - 2010]

Sol: $f^1(x) = \begin{vmatrix} -2 \sin x & 0 & 0 \\ x - \frac{\pi}{2} & 2 \cos x & 1 \\ 0 & 1 & 2 \cos x \end{vmatrix} +$

$$\begin{vmatrix} 2 \cos x & 1 & 0 \\ 1 & -2 \sin x & 0 \\ 0 & 1 & 2 \cos x \end{vmatrix} +$$

$$\begin{vmatrix} 2 \cos x & 1 & 0 \\ x - \frac{\pi}{2} & 2 \cos x & 1 \\ 0 & 0 & -2 \sin x \end{vmatrix}$$

put $x = \pi$ we get $f^1(\pi) = 2$

Parametric Differentiation:

- If $x=f(t)$ and $y=g(t)$ are the parametric equations of a curve then

i) $\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{g^1(t)}{f^1(t)}$

ii) $\frac{d^n y}{dx^n} = \frac{d}{dt} \left(\frac{d^{n-1} y}{dx^{n-1}} \right) \left(\frac{dt}{dx} \right)$

W.E-6: If $x = a \cos^3 \theta$, $y = a \sin^3 \theta$ then $\frac{dy}{dx} =$

Sol: $\frac{dx}{d\theta} = -3a \cos^2 \theta \cdot \sin \theta$

and $\frac{dy}{d\theta} = 3a \sin^2 \theta \cdot \cos \theta$

then $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = -\tan \theta$

Derivative of Implicit Functions:

- If $f(x, y) = 0$, then differentiate each term w.r.t. x regarding y as a function of x and then collect the terms of $\frac{dy}{dx}$ together on left hand side and remaining terms on

right handside and then find $\frac{dy}{dx}$.

Alternative method : If $f(x, y) = 0$

$$\text{then } \frac{dy}{dx} = - \frac{\left(\frac{\partial f}{\partial x} \right)}{\left(\frac{\partial f}{\partial y} \right)}$$

$$= - \frac{\text{Partial derivative of } f \text{ w.r.t. } x}{\text{Partial derivative of } f \text{ w.r.t. } y}$$

W.E-7: If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$,

$$\text{then } \frac{dy}{dx} = \frac{-(ax + hy + g)}{hx + by + f}$$

Logarithmic Differentiation:

➤ If $y = \{f_1(x)\}^{f_2(x)}$ or
 $y = f_1(x) \cdot f_2(x) \cdot f_3(x) \dots$ or

$$y = \frac{f_1(x) \cdot f_2(x) \cdot f_3(x) \dots}{g_1(x) \cdot g_2(x) \cdot g_3(x) \dots}$$

then take logarithm on both sides and differentiate both sides w.r.t. x

W.E-8: $x^y = y^x$ then $\frac{dy}{dx} =$

Sol: Take logarithm, we get, $y \log x = x \log y$
 differentiating w.r.t x , we get

$$\log x \frac{dy}{dx} + y \cdot \frac{1}{x} = 1 \cdot \log y + x \cdot \frac{1}{y} \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{y}{x} \cdot \frac{(x \log y - y)}{(y \log x - x)}$$

Derivative of Composite Function:

➤ $y = (g \circ f)(x) \Rightarrow y = g(f(x))$
 $\Rightarrow \frac{dy}{dx} = g^1(f(x)) \cdot f^1(x)$

W.E-9: If $f(x) = e^x$, $g(x) = \sin^{-1}x$ and

$$h(x) = f(g(x)), \text{ then } \frac{h^1(x)}{h(x)} =$$

Sol: $h^1(x) = f^1(g(x))g^1(x)$

$$\Rightarrow \frac{h^1(x)}{h(x)} = \frac{f^1(g(x))g^1(x)}{f(g(x))} = \frac{1}{\sqrt{1-x^2}}$$

Standard Derivatives:

➤ i) If $y = \sqrt[n]{f(x) + \sqrt[n]{f(x) + \sqrt[n]{f(x) + \dots \text{to } \infty}}$ then

$$\frac{dy}{dx} = \frac{f^1(x)}{ny^{n-1} - 1}$$

ii) If $y = \sqrt{f(x) + \sqrt{f(x) + \sqrt{f(x) + \dots \text{to } \infty}}$ then

$$\frac{dy}{dx} = \frac{f^1(x)}{(2y-1)}$$

iii) Let $f(x)$, $g(x)$ be the two differentiable functions of 'x' and if $y = f(x)^{g(x)}$ then

$$\frac{dy}{dx} = y \left(g^1(x) \log f(x) + g(x) \frac{f^1(x)}{f(x)} \right)$$

iv) If $y = f(x) + \frac{1}{y}$ then $\frac{dy}{dx} = \frac{y^2 f^1(x)}{y^2 + 1}$

v) If $f(x+y) = f(x) \cdot f(y) \quad \forall x, y \in \mathbb{R}$ and $f(x) \neq 0$
 then $f^1(x) = f^1(0) \cdot f(x)$.

W.E-10: If $f(x+y) = f(x) \cdot f(y) \quad \forall x \in \mathbb{R}$,

$$f(5)=2, f^1(0)=3 \text{ then } f^1(5)=$$

Sol: $f^1(5) = f(5) \cdot f^1(0) = 2(3) = 6$.

vi) If $f(x) = |x|$, then $f^1(0)$ does not exist.

$$\text{vii) } \frac{d}{dx} \left(\frac{ax+b}{cx+d} \right) = \frac{ad-bc}{(cx+d)^2}$$

$$\text{viii) } \frac{d}{dx} \left(\frac{af(x)+b}{cf(x)+d} \right) = \frac{(ad-bc)f^1(x)}{\{cf(x)+d\}^2}$$

$$\text{ix) } \frac{d}{dx} \{ \log_e f(x) \} = \frac{f^1(x)}{f(x)}$$

$$\text{x) } \frac{d}{dx} \{ \sqrt[n]{f(x)} \} = \frac{f^1(x)}{2\sqrt[n]{f(x)}}$$

$$\text{xi) } \frac{d}{dx} \left\{ \frac{1}{f(x)} \right\} = \frac{-f^1(x)}{(f(x))^2}$$

$$\text{xii) } \frac{d}{dx} \{f(x)^n\} = n \{f(x)\}^{n-1} \cdot f'(x)$$

$$\text{xiii) If } y = \tan^{-1} \left(\frac{f(x) \pm g(x)}{1 \mp f(x) \cdot g(x)} \right) \text{ then}$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + \{f(x)\}^2} \pm \frac{g'(x)}{1 + \{g(x)\}^2}$$

$$\text{xiv) If } y = f(x) \text{ and } z = g(x) \text{ then}$$

$$\frac{dy}{dz} = \frac{f'(x)}{g'(x)}$$

$$\text{xv) If } y = f(x)^y \text{ then}$$

$$\frac{dy}{dx} = \frac{y^2 \cdot f'(x)}{f(x) \{1 - y \log_e f(x)\}}$$

Substitutions:

➤ While differentiating the given function using trigonometric transformation, observe the following points.

i) If the function involve the term $\sqrt{a^2 - x^2}$,

then put $x = a \sin \theta$ (or) $x = a \cos \theta$

ii) If the function involve the term $\sqrt{a^2 + x^2}$,

then put $x = a \tan \theta$ (or) $x = a \cot \theta$

iii) If the function involve the term $\sqrt{x^2 - a^2}$,

then put $x = a \sec \theta$ (or) $x = a \csc \theta$

iv) If the function involve the term

$$\sqrt{\frac{a-x}{a+x}} \text{ (or) } \sqrt{\frac{a+x}{a-x}}, \text{ then}$$

put $x = a \cos \theta$ (or) $x = a \cos 2\theta$

W.E-11: $y = \tan^{-1} \left(\frac{x}{\sqrt{1+x^2}-1} \right)$, then $\frac{dy}{dx} =$

Sol: Put $x = \tan \theta$ then

$$y = \tan^{-1} \left(\cot \frac{\theta}{2} \right)$$

$$= \frac{\pi}{2} - \frac{1}{2} \tan^{-1} x$$

$$\therefore \frac{dy}{dx} = -\frac{1}{2(1+x^2)}$$

Higher Order Derivatives of functions:

➤ i) $\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right)$

ii) $\frac{d^3 y}{dx^3} = \frac{d}{dx} \left(\frac{d^2 y}{dx^2} \right)$

W.E-12: If $y = \sin(\log_e x)$, then

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} = \quad \quad \quad \text{(EAM-2008)}$$

Sol: $\frac{dy}{dx} = \frac{\cos(\log x)}{x} \Rightarrow x \frac{dy}{dx} = \cos(\log_e x)$

$$\Rightarrow x \frac{d^2 y}{dx^2} + \frac{dy}{dx} = \frac{-\sin(\log_e x)}{x}$$

$$\Rightarrow x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} = -y$$

I. The derivative of an even function is odd and that of an odd function is even

II. If 'f' is even then $f'(0) = 0$

III. If $f(x)$ is a periodic function which is differentiable everywhere then its derivative is also periodic.

IV. If a differentiable function is symmetric about the vertical line $x=a$ then $f'(a) = 0$

V. To find the derivatives of functions involving inverse trigonometric functions we use following substitutions.

Function	Substitution
a) $\sqrt{(a-x)(x-b)}$	$x = a \cos^2 \theta + b \sin^2 \theta$
b) $\sqrt{2a-x}$	$x = 2a \sin \theta$ or $x = 2a \cos \theta$
c) $\sqrt{x^4 + a^4}$	$x^2 = a^2 \tan \theta$

d) $\sqrt{x^4 - a^4} \quad x^2 = a^2 \sec \theta$

e) $\sqrt{2ax - x^2} \quad x = 2a \sin^2 \theta$

VI. a) If $f\left(\frac{y}{x}\right) = C$ then $\frac{dy}{dx} = \frac{y}{x}$

b) If $f(xy) = C$ then $\frac{dy}{dx} = -\frac{y}{x}$

c) If $f\left(\frac{x}{y}\right) = C$ then $\frac{dy}{dx} = \frac{y}{x}$

d) If $f(x^m y^n) = C$ then $\frac{dy}{dx} = -\frac{my}{nx}$

(C is constant)

W.E - 13 : If $y = x \sin^{-1}\left(\frac{x}{y}\right)$ then $\frac{dy}{dx} =$

Ans. $\frac{y}{x}$

Sol:

$$\left(\frac{x}{y}\right) \sin^{-1}\left(\frac{x}{y}\right) = 1 \Rightarrow \text{from VI (c), } \frac{dy}{dx} = \frac{y}{x}$$

W.E - 14 : If $xy = (x+y)^n$ ($n \in \mathbb{N}$) and $\frac{dy}{dx} = \frac{y}{x}$

then $n =$

Ans. 2

Sol: It must be homogeneous function

$$\therefore n = 1 + 1 = 2$$

W.E - 15: If $x^{21} \cdot y + x^7 \cdot y^{15} + x^8 \cdot y^{14} + x \cdot y^{21}$

$= a(x^{22} + y^{22})$ and a is such that $(1, 2)$ is a point on the curve then $\frac{dy}{dx}$ at $(1, 2)$ is

Ans. 2

Sol: Given equation takes the form $f\left(\frac{y}{x}\right) = C$
 $\Rightarrow \frac{dy}{dx} = \frac{y}{x} \Rightarrow \left(\frac{dy}{dx}\right)_{(1,2)} = 2$

W.E - 16 : If $f(xy) + g(xy) = K$ then $\frac{dy}{dx} =$

Ans. $-\frac{y}{x}$

Sol: $f(xy) = C \Rightarrow \frac{dy}{dx} = -\frac{y}{x}$

(It can also be found by using implicit differentiation)



LEVEL - I (C.W)

Formula Based:

1. If $y = (\log x)^3$ then $\frac{dy}{dx} =$

1) $\frac{3(\log x)^2}{x}$ 2) $3(\log x)^2$

3) $\frac{(\log x)^2}{x}$ 4) $(\log x)^2$

2. $\frac{d}{dx}(\sqrt{\sin \sqrt{x}}) =$

1) $\frac{\cos \sqrt{x}}{4\sqrt{x \sin \sqrt{x}}}$ 2) $\frac{\sin \sqrt{x}}{4\sqrt{x \cos \sqrt{x}}}$

3) $\frac{-\cos \sqrt{x}}{4\sqrt{x \sin \sqrt{x}}}$ 4) $\frac{-\tan \sqrt{x}}{4\sqrt{x \cos \sqrt{x}}}$

3. If $y = a^{a^x}$ then $\frac{dy}{dx} =$

1) $y \cdot a^x (\log a)^2$ 2) $y \cdot a^x \cdot \log a$
 3) $(y \cdot a^x)^2$ 4) $(y \cdot a^x)$

4. $\frac{d}{dx}[\log\{\log(\log x)\}] =$

1) $\frac{1}{x \log x \log(\log x)}$ 2) $\frac{-1}{x \log x \log(\log x)}$

$$3) \frac{x}{\log x \log(\log x)} \quad 4) \frac{1}{\log x \log(\log x)}$$

Inverse Trigonometric Functions and Substitutions:

5. If $y = 2^{ax}$ and $\frac{dy}{dx} = \log 256$ at $x=1$, then the value of a is

$$1) 0 \quad 2) 1 \\ 3) 2 \quad 4) 3$$

6. If $y = \tan^{-1}\left(\cot\left(\frac{\pi}{2} - x\right)\right)$ then $\frac{dy}{dx} =$

$$1) 1 \quad 2) -1 \quad 3) 0 \quad 4) \frac{1}{2}$$

7. If $y = \tan^{-1}(\sec x + \tan x)$ then $\frac{dy}{dx} =$

$$1) 1 \quad 2) \frac{1}{2} \quad 3) -1 \quad 4) 0$$

8. If $y = \tan^{-1}\left(\frac{(3-x)\sqrt{x}}{1-3x}\right)$ then $\frac{dy}{dx} =$

$$1) \frac{3}{(1+x)\sqrt{x}} \quad 2) \frac{3}{2(1+x)\sqrt{x}} \\ 3) \frac{-3}{2(1+x)\sqrt{x}} \quad 4) 0$$

Logarithmic Differentiation:

9. If $y = (\log x)^x$ then $\frac{dy}{dx} =$

$$1) (\log x)^x \left[\log(\log x) + \frac{1}{\log x} \right] \\ 2) (\log x)^x \left[\log(\log x) - \frac{1}{\log x} \right] \\ 3) -(\log x)^x \left[\log(\log x) + \frac{1}{\log x} \right] \\ 4) -(\log x)^x \left[\log(\log x) - \frac{1}{\log x} \right]$$

10. If $y = (\sin x)^x$ then $\frac{dy}{dx} =$

$$1) y(\log(\sin x) + x \cot x) \\ 2) y(\log(\sin x) - x \cot x) \\ 3) -y(\log(\sin x) - x \cot x) \\ 4) -y(\log(\sin x) + x \cot x)$$

11. If $x^y = e^{x-y}$ then $\frac{dy}{dx} =$

$$1) \frac{\log x}{(1+\log x)^2} \quad 2) -\frac{\log x}{(1+\log x)^2} \\ 3) -\frac{\log x}{(1-\log x)^2} \quad 4) \frac{\log x}{(1-\log x)^2}$$

Derivative of Hyperbolic and Inverse Functions:

12. $\frac{d}{dx} [\log(\cosh x)] =$

$$1) \sinh x \quad 2) \coth x \\ 3) \tanh x \quad 4) -\coth x$$

13. $\frac{d}{dx} (\operatorname{cosech}(x^3)) =$

$$1) -\operatorname{cosech}(x^3) \cdot \coth(x^3) \cdot (3x^2) \\ 2) \operatorname{cosech}(x^3) \cdot \coth(x^3) \cdot (3x^2) \\ 3) \operatorname{cosech}(x^3) \cdot \coth(x^3) \cdot (6x^2) \\ 4) -\operatorname{cosech}(x^3) \cdot \coth(x^3) \cdot (6x^2)$$

14. $\frac{d}{dx} (\sinh^{-1}(3x)) =$

$$1) \frac{1}{\sqrt{1+9x^2}} \quad 2) \frac{2}{\sqrt{1+9x^2}} \\ 3) \frac{3}{\sqrt{1+9x^2}} \quad 4) \frac{9}{\sqrt{1+9x^2}}$$

Parametric Differentiation:

15. If $x=a(t+\sin t)$, $y=a(1-\cos t)$ if $\frac{dx}{dy} = \cot p$ then $p =$

- 1) t 2) $2t$ 3) $\frac{t}{2}$ 4) $-t^2$
16. If $x = a(\cos t + \log(\tan \frac{t}{2}))$, $y = a \sin t$ then $\frac{dy}{dx} =$ (EAM-2008)

1) $\sin t$ 2) $\cot t$ 3) $\tan t$ 4) $\tan^2 t$

17. If $x = a \cos^4 t$, $y = b \sin^4 t$ then $\frac{dy}{dx}$ at $t = \frac{3\pi}{4}$ is
- 1) $\frac{-b}{a}$ 2) $\frac{b}{a}$ 3) $\frac{a}{b}$ 4) $\frac{-a}{b}$

Derivative of one function w.r.t. another function:

18. The derivative of a^x w.r.t. $\sin^{-1} x$ is

1) $a^x \cdot \log_e a \sqrt{1-x^2}$ 2) $\frac{a^x \cdot \log_e a}{\sqrt{1-x^2}}$

3) $\frac{a^x}{\sqrt{1-x^2}}$ 4) $\frac{-a^x}{\sqrt{1-x^2}}$

19. The derivative of $\cos^{-1} \frac{1-x^2}{1+x^2}$ w.r.t.

$\tan^{-1} \frac{2x}{1-x^2}$ is

1) 0 2) 1 3) 2 4) $\frac{1}{2}$

20. Derivative of $\sin^{-1} x$ with respect to $\cos^{-1} \sqrt{1-x^2}$

1) $\frac{1}{\sqrt{1-x^2}}$ 2) $\cos^{-1} x$ 3) 1 4) 0

Derivative of implicit function:

21. If $2x^2 - 3xy + y^2 + x + 2y - 8 = 0$, then $\frac{dy}{dx}$ is equal to (EAM-2007)

1) $\frac{3y-4x-1}{2y-3x+2}$ 2) $\frac{3y+4x+1}{2y+3x+2}$

3) $\frac{3y-4x+1}{2y-3x-2}$ 4) $\frac{3y-4x+1}{2y+3x+2}$

22. If $x + y = \sin(x + y)$ then $\frac{dy}{dx} =$

1) $\frac{1}{2}$ 2) 0 3) -1 4) $\frac{1}{3}$

23. If $\sin(x + y) = \log(x + y)$, then $\frac{dy}{dx} =$

1) 2 2) -2 3) 1 4) -1

LEVEL-I (C.W)-KEY

1) 1 2) 1 3) 1 4) 1 5) 3
6) 1 7) 2 8) 2 9) 1 10) 1
11) 1 12) 3 13) 1 14) 3 15) 3
16) 3 17) 1 18) 1 19) 2 20) 3
21) 1 22) 3 23) 4

LEVEL-I (C.W)-HINTS

- $3(\log x)^2 \cdot \frac{1}{x}$
- $\frac{1}{2\sqrt{\sin \sqrt{x}}} \cos \sqrt{x} \cdot \frac{1}{2\sqrt{x}}$
- $a^x \log a \cdot \frac{d}{dx}(x)$
- $\frac{1}{\log(\log x)} \cdot \frac{1}{\log x} \cdot \frac{1}{x}$
- Take logarithm and differentiate
- $y = \tan^{-1}(\tan x) = x$
- $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$
- Put $\sqrt{x} = \tan \theta$
- Take logarithm on both sides & differentiate.
- Take logarithm on both sides & differentiate.
- Taking logarithm on both sides & differentiate
- $\frac{1}{\cosh x} \cdot \sinh x$
- $-\operatorname{cosech}(x^3) \cdot \cot h(x^3) \cdot (3x^2)$
- $\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1+x^2}}$

$$15. \frac{dx}{dy} = \frac{\left(\frac{dx}{dt}\right)}{\left(\frac{dy}{dt}\right)} = \cot p$$

$$16. \frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)}$$

$$17. \frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)} \text{ and substitute } t = 3\pi/4$$

$$18. \text{ Let } y = a^x, z = \sin^{-1} x, \frac{dy}{dz} = \frac{\left(\frac{dy}{dx}\right)}{\left(\frac{dz}{dx}\right)}$$

$$19. \text{ Put } x = \tan \theta$$

$$20. y = \sin^{-1} x, z = \cos^{-1} \sqrt{1-x^2} \quad (x = \sin \theta)$$

$$21. \frac{dy}{dx} = \frac{-\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}$$

$$22. \frac{dy}{dx} = \frac{-\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}}$$

$$23. \cos(x+y) \left(1 + \frac{dy}{dx}\right) = \frac{1}{x+y} \left(1 + \frac{dy}{dx}\right)$$



Formula Based:

$$1. \frac{d}{dx} \left\{ e^{\log \sqrt{1+\cot^2 x}} \right\} =$$

- 1) cosec x cot x 2) -cosec x.cot x
3) cosec² x.cot x 4) 0

$$2. \frac{d}{dx} (\sin \sqrt{2x-3}) =$$

- 1) $\frac{\cos \sqrt{2x-3}}{\sqrt{2x-3}}$ 2) $\frac{-\cos \sqrt{2x-3}}{2\sqrt{2x-3}}$
3) $\sqrt{2x-3} \cos \sqrt{2x-3}$ 4) $\cos \sqrt{2x-3}$

$$3. \text{ If } f(x) = \frac{a^x}{x^a} \text{ then } f'(a) =$$

- 1) $\log a - 1$ 2) $\log a - a$
3) $a \log a - a$ 4) $a \log a + a$

$$4. \frac{d}{dx} \{ x^n \cdot \log x \} =$$

- 1) $x^n (1 + n \log x)$ 2) $x^{n-1} (1 + n \log x)$
3) $x^{n-1} (1 - n \log x)$ 4) $x^{n-1} (1 - n \log x)$

Inverse Trigonometric Functions and Substitutions:

$$5. \text{ If } y = \tan^{-1} \left(\frac{\cos x}{1 + \sin x} \right) \text{ then } \frac{dy}{dx} =$$

- 1) 1 2) -1 3) $-\frac{1}{2}$ 4) $\frac{1}{3}$

$$6. \frac{d}{dx} \left\{ \tan^{-1} \frac{x}{1+x^2} + \tan^{-1} \frac{1+x^2}{x} \right\} =$$

- 1) 0 2) 1 3) $\frac{1}{2}$ 4) 2

$$7. \text{ If } y = \tan^{-1} \left(\frac{3a^2x - x^3}{a^3 - 3ax^2} \right) \text{ then } \frac{dy}{dx} =$$

- 1) $\frac{3a}{a^2 + x^2}$ 2) $\frac{1}{a^2 + x^2}$
3) $\frac{-3a^2}{a^2 + x^2}$ 4) $\frac{-3a}{a^2 + x^2}$

Logarithmic Differentiation:

$$8. \text{ If } y = 2^{2^x} \text{ then } \frac{dy}{dx} =$$

- 1) $y \cdot (\log 2)^2 \cdot 2^x$ 2) $y (\log 2) \cdot 2^x$

3) $y2(\log 2)^2 2^x$

4) $-y(\log 2) \cdot 2^x$

9. If $x^m \cdot y^n = (x+y)^{m+n}$ then $\frac{dy}{dx} =$

1) $\frac{y}{x}$ 2) $-\frac{y}{x}$ 3) $\frac{my}{x}$ 4) $\frac{ny}{x}$

10. If $y = x^{\cos x}$ then $\frac{dy}{dx} =$

1) $x^{\cos x} \left(\frac{\cos x}{x} + \sin x \cdot \log x \right)$

2) $x^{\cos x} \left(\frac{\cos x}{x} - \sin x \cdot \log x \right)$

3) $-x^{\cos x} \left(\frac{\cos x}{x} - \sin x \cdot \log x \right)$

4) $-x^{\cos x} \left(\frac{\cos x}{x} + \sin x \cdot \log x \right)$

Derivative of Hyperbolic and Inverse Functions:

11. $\frac{d}{dx} \left(\cosh^{-1} \frac{x}{2} \right) =$

1) $\frac{1}{\sqrt{x^2+4}}$

2) $\frac{1}{\sqrt{x^2-4}}$

3) $\frac{-1}{\sqrt{x^2+4}}$

4) $\frac{-1}{\sqrt{x^2-4}}$

12. $\frac{d}{dx} \left(\sinh^{-1} (\sqrt{x}) \right) =$

1) $\frac{1}{\sqrt{1+x}} \cdot \frac{1}{\sqrt{x}}$

2) $\frac{1}{\sqrt{1+x}} \cdot \frac{1}{2\sqrt{x}}$

3) $-\frac{1}{\sqrt{1+x}} \cdot \frac{1}{2\sqrt{x}}$

4) $-\frac{1}{\sqrt{1+x}} \cdot \frac{1}{\sqrt{x}}$

13. $\frac{d}{dx} (\operatorname{sech}(\log x)), (x > 0)$

1) $-\operatorname{sech}(\log x) \cdot \tanh(\log x) \cdot \frac{1}{x}$

2) $\operatorname{sech}(\log x) \cdot \tanh(\log x) \cdot \frac{1}{x}$

3) $\operatorname{sech}(\log x) \cdot \coth(\log x) \cdot \frac{1}{x}$

4) $-\operatorname{sech}(\log x) \cdot \coth(\log x) \cdot \frac{1}{x}$

Parametric Differentiation:

14. If $x = at^2, y = 2at$ then $\frac{dy}{dx} =$

1) $1/t$ 2) t 3) t^2 4) 1

15. If $x = a(1 - \cos \theta), y = a(\theta + \sin \theta)$ then $\frac{dy}{dx}$ at

$\theta = \frac{\pi}{2}$ is

1) 1 2) $\frac{1}{2}$ 3) 0 4) $-\frac{1}{2}$

16. If $x = a \cos^4 t, y = b \sin^4 t$ then $\frac{dy}{dx}$ at $t = \frac{\pi}{4}$ is

1) $-\frac{b}{a}$ 2) $\frac{b}{a}$ 3) $\frac{a}{b}$ 4) $-\frac{a}{b}$

Derivative of one function w.r.t. another function:

17. The derivative of $e^{\sin^{-1} x}$ w.r.t. $\log x$ is

1) $\frac{-xe^{\sin^{-1} x}}{\sqrt{1-x^2}}$

2) $\frac{xe^{\sin^{-1} x}}{\sqrt{1-x^2}}$

3) $\frac{e^{\sin^{-1} x}}{\sqrt{1-x^2}}$

4) $e^{\sin^{-1} x} \cdot \sqrt{1-x^2}$

18. The derivative of $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$ w.r.t. $\tan^{-1} x$ is

1) 0 2) 1 3) 2 4) $\frac{1}{2}$

19. The derivative of $\cos^{-1}(2x^2-1)$ with respect to $\cos^{-1} x$ is

1) 2 2) $\frac{2}{x}$

3) $\frac{1}{2\sqrt{1-x^2}}$ 4) $\sqrt{1-x^2}$

Derivative of implicit function:**LEVEL - II(C.W)**

20. If $ax^2 + 2hxy + by^2 = 0$ then $\frac{dy}{dx} =$

1) $-\left(\frac{ax+hy}{hx+by} \right)$

2) $\left(\frac{ax+hy}{hx+by} \right)$

3) $-(ax+hy)(hx+by)$ 4) $(ax+hy)(hx+by)$

21. If $x^2 - y^2 = a(x - y)$ and $x \neq y$, then $\frac{dy}{dx} =$

- 1) 1 2) $\frac{1}{2}$ 3) $\frac{1}{3}$ 4) -1

22. If $e^{x+y} = xy$ then $\frac{dy}{dx} =$

- 1) $\frac{y(1-x)}{x(y-1)}$ 2) $\frac{-y(1-x)}{x(y-1)}$ 3) $\frac{x(y-1)}{y(1+x)}$ 4) $\frac{-x(y-1)}{y(1+x)}$

LEVEL-I (H.W)-KEY

- 1) 2 2) 1 3) 1 4) 2 5) 3
 6) 1 7) 1 8) 1 9) 1 10) 2
 11) 2 12) 2 13) 1 14) 1 15) 1
 16) 1 17) 2 18) 4 19) 1 20) 1
 21) 4 22) 1

LEVEL-I (H.W)-HINTS

- $\frac{d}{dx}(e^{\log_e \csc x}) = \frac{d}{dx}(\csc x)$
- $\cos(\sqrt{2x-3}) \cdot \frac{1}{2\sqrt{2x-3}}$
- $\left(\frac{u}{v}\right)' = \frac{vu' - uv'}{v^2}$
- $(uv)' = uv' + vu'$
- $y = \tan^{-1}\left(\frac{\sin(\frac{\pi}{2} - x)}{1 + \cos(\frac{\pi}{2} - x)}\right)$
- $\frac{d}{dx}(\tan^{-1} x + \cot^{-1} x) = \frac{d}{dx}\left(\frac{\pi}{2}\right) = 0$
- Put $x = a \tan \theta$
- $\frac{d}{dx}(a^x) = a^x \log a \cdot \frac{d}{dx}(x)$
- Take logarithm both sides and differentiate
- Take logarithm both sides and differentiate
- $\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2 - 1}}$
- $\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1 + x^2}}$

13. $\frac{d}{dx}(\sec hx) = -(\sec hx) \cdot (\tan hx)$

14. $\frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)}$

15. $\frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)}$ and substitute $\theta = \frac{\pi}{2}$

16. $\frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)}$

17. $y = e^{\sin^{-1} x}$, $z = \log x$ and $\frac{dy}{dz} = \frac{\left(\frac{dy}{dx}\right)}{\left(\frac{dz}{dx}\right)}$

18. $\frac{dy}{dz} = \frac{\left(\frac{dy}{dx}\right)}{\left(\frac{dz}{dx}\right)}$, put $x = \tan \theta$

19. $y = \cos^{-1}(2x^2 - 1)$, $z = \cos^{-1} x$ and

$\frac{dy}{dz} = \frac{\left(\frac{dy}{dx}\right)}{\left(\frac{dz}{dx}\right)}$

20. $\frac{dy}{dx} = \frac{-\left(\frac{\partial f}{\partial x}\right)}{\left(\frac{\partial f}{\partial y}\right)}$

21. $(x + y)(x - y) = a(x - y) \Rightarrow x + y = a$

22. $e^x e^y = xy$ and $\frac{dy}{dx} = \frac{-\left(\frac{\partial f}{\partial x}\right)}{\left(\frac{\partial f}{\partial y}\right)}$



Formula Based:

1. $\frac{d}{dx}(\sin^5 x \cdot \sin 5x) =$

- 1) $\sin^4 x \cdot \sin 5x$ 2) $5\sin^4 x \cdot \sin 6x$
 3) $5\sin^4 x \cdot \sin 5x$ 4) $-5\sin^4 x \cdot \sin 6x$

2. If $y = \sqrt{\frac{1-\sin 2x}{1+\sin 2x}}$ then $\left(\frac{dy}{dx}\right)_{x=0} =$
 1) $\frac{1}{2}$ 2) 1 3) -2 4) 2
3. If $y = e^{\sin^2 x + \sin^4 x + \sin^6 x + \dots + \infty}$, then $\frac{dy}{dx} =$
 1) $e^{\tan^2 x}$ 2) $e^{\tan^2 x} \sec^2 x$
 3) $2e^{\tan^2 x} \tan x \sec^2 x$ 4) 1
4. If $y = x + e^x$, then $\frac{d^2x}{dy^2}$ is equal to
 1) $\frac{1}{(1+e^x)^2}$ 2) $\frac{-e^x}{(1+e^x)^2}$
 3) $\frac{-e^x}{(1+e^x)^3}$ 4) e^x
5. If $x \cdot e^{xy} = y + \sin^2 x$. then at $x = 0$, $\frac{dy}{dx} =$
 1) 1 2) 2 3) 3 4) 0
- Inverse Trigonometric Functions and Substitutions:**
6. If $y = \tan^{-1}\left(\frac{2^x}{1+2^{2x+1}}\right)$ then $\frac{dy}{dx}$ at $x = 0$ is
 1) $\frac{1}{10} \log 2$ 2) $\frac{1}{5} \log 2$
 3) $-\frac{1}{10} \log 2$ 4) $\log 2$
7. If $y = \sec(\tan^{-1} x)$, then $\frac{dy}{dx}$ at $x = 1$ is equal to
 1) $\frac{1}{\sqrt{2}}$ 2) $\frac{1}{2}$ 3) 1 4) $\sqrt{2}$
8. $\frac{d}{dx} \left(\tan^{-1} \frac{x}{\sqrt{a^2 - x^2}} \right) =$
 1) $\frac{1}{\sqrt{a^2 + x^2}}$ 2) $\frac{-1}{\sqrt{a^2 + x^2}}$
 3) $\frac{x}{\sqrt{a^2 - x^2}}$ 4) $\frac{1}{\sqrt{a^2 - x^2}}$

9. $\frac{d}{dx} \left\{ \cot^{-1} \frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}} \right\} =$
 1) $\frac{1}{\sqrt{1-x^2}}$ 2) $\frac{-1}{2\sqrt{1-x^2}}$ 3) $\frac{1}{1+x^2}$ 4) $\frac{-1}{2(1+x^2)}$
10. $\frac{d}{dx} \left(\sin^{-1} \left\{ \frac{\sqrt{1+x} + \sqrt{1-x}}{2} \right\} \right) =$
 1) $\frac{-1}{2\sqrt{1-x^2}}$ 2) $\frac{1}{2\sqrt{1-x^2}}$ 3) $\frac{1}{\sqrt{1-x^2}}$ 4) $\frac{-1}{\sqrt{1-x^2}}$
11. If $y = \cos^{-1} \left(\frac{x-x^{-1}}{x+x^{-1}} \right)$ then $\frac{dy}{dx} =$
 1) $\frac{-2}{1+x^2}$ 2) $\frac{2}{1+x^2}$ 3) $\frac{1}{1+x^2}$ 4) $\frac{-1}{1+x^2}$

Derivative of Hyperbolic and Inverse Functions:

12. $\frac{d}{dx} \left(\frac{3^x - 3^{-x}}{3^x + 3^{-x}} \right) =$
 1) $\frac{4 \log 3}{(3^x + 3^{-x})^2}$ 2) $\frac{-4 \log 3}{(3^x + 3^{-x})^2}$
 3) $\frac{1}{(3^x + 3^{-x})^2}$ 4) $\frac{-1}{(3^x + 3^{-x})^2}$
13. $\frac{d}{dx} (\tan^{-1}(\sinh x)) =$
 1) cosechx 2) sechx
 3) sinhx 4) coshx

Logarithmic Differentiation:

14. $\frac{d}{dx} \{ x^{\tan x} \} =$
 1) $x^{\tan x} \left\{ \frac{\tan x}{x} + \sec^2 x \cdot \log x \right\}$
 2) $x^{\tan x} \left\{ \frac{\tan x}{x} - \sec^2 x \cdot \log x \right\}$
 3) $x^{\tan x} \left\{ \frac{x}{\tan x} + \sec^2 x \cdot \log x \right\}$
 4) $x^{\tan x} \left\{ \frac{x}{\tan x} - \sec^2 x \cdot \log x \right\}$

26. If $y = \sqrt{a^{3x+1} + \sqrt{a^{3x+1} + \sqrt{a^{3x+1} + \dots \text{to } \infty}}}$

then $\frac{dy}{dx} =$

1) $\frac{a^{3x+1} \cdot \log a}{(2y-1)}$ 2) $\frac{3 \cdot a^{3x+1} \cdot \log a}{(2y-1)}$

3) $\frac{3 \cdot a^{3x+1} \cdot \log a}{(2y+1)}$ 4) $\frac{a^{3x+1} \cdot \log a}{(2y+1)}$

27. If $x\sqrt{1+y} + y\sqrt{1+x} = 0$ and $x \neq y$ then $\frac{dy}{dx} =$

1) $\frac{x+1}{(1-x)^2}$ 2) $\frac{1}{(1-x)^2}$

3) $\frac{-1}{(1+x)^2}$ 4) $\frac{1}{(1+x)^2}$

Derivative of composite function:

28. If $f(x) = \frac{x}{\sqrt{1-x^2}}$, $g(x) = \frac{x}{\sqrt{1+x^2}}$ then

$\frac{d}{dx}(f \circ g(x)) =$

1) 1 2) 0 3) -1 4) 2

LEVEL-II (C.W)-KEY

- 1) 2 2) 3 3) 3 4) 3 5) 1
6) 3 7) 1 8) 4 9) 2 10) 1
11) 1 12) 1 13) 2 14) 1 15) 1
16) 4 17) 3 18) 1 19) 2 20) 4
21) 1 22) 1 23) 2 24) 1 25) 1
26) 2 27) 3 28) 1

LEVEL-II (C.W)-HINTS

1. $\frac{d}{dx}(\sin^n x \cdot \sin nx) = n \sin^{n-1} x (\sin(n+1)x)$

2. $y = \sqrt{\frac{1 - \cos\left(\frac{\pi}{2} - 2x\right)}{1 + \cos\left(\frac{\pi}{2} - 2x\right)}}$

3. $y = e^{\sin^2 x (1 + \sin^2 x + \sin^4 x + \dots + \infty)} = e^{\sin^2 x (1 - \sin^2 x)^{-1}}$
 $= e^{\left(\frac{\sin^2 x}{\cos^2 x}\right)} = e^{\tan^2 x}$

4. $\Rightarrow \frac{dy}{dx} = 1 + e^x$

$\Rightarrow \frac{dx}{dy} = \frac{1}{1 + e^x}$

$\Rightarrow \frac{d^2 x}{dy^2} = \frac{d}{dy} \left(\frac{1}{1 + e^x} \right)$

$\Rightarrow \frac{d^2 x}{dy^2} = \frac{-1}{(1 + e^x)^2} \frac{d}{dy} (1 + e^x)$

$\Rightarrow \frac{d^2 x}{dy^2} = \frac{-1}{(1 + e^x)^2} e^x \frac{dx}{dy} = \frac{-e^x}{(1 + e^x)^3}$

5. If $x = 0$ then $y = 0$

$e^{xy} + x \cdot e^{xy} \left[x \cdot \frac{dy}{dx} + y \right] = \frac{dy}{dx} + 2 \sin x \cos x$

Put $x = 0$, $y = 0$, we get $\left(\frac{dy}{dx} \right)_{x=0} = 1$

6. $y = \tan^{-1} \left(\frac{2^{x+1} - 2^x}{1 + 2^x \cdot 2^{x+1}} \right)$

7. Given, $y = \sec(\tan^{-1} x)$

Let $\tan^{-1} x = \theta$

$\Rightarrow x = \tan \theta$

$\therefore y = \sec \theta = \sqrt{1 + x^2}$

differentiate w.r.t. x .

8. $x = a \sin \theta$

LEVEL - II (H.W)

9. Put $x = \cos \theta$

10. Put $x = \cos \theta$

11. Put $x = \tan \theta$

12. $\frac{d}{dx} \left(\frac{e^{\log_e 3^x} - e^{\log_e 3^{-x}}}{e^{\log_e 3^x} + e^{\log_e 3^{-x}}} \right)$

13. $\frac{d}{dx} (\tan^{-1} x) = \frac{1}{1 + x^2}$

$$14. \frac{d}{dx} \{f(x)^{g(x)}\} =$$

$$f(x)^{g(x)} \left\{ g(x) \cdot \frac{1}{f(x)} \cdot f'(x) + \log f(x) \cdot g'(x) \right\}$$

$$15. \frac{d}{dx} \{f(x)^{g(x)}\} =$$

$$f(x)^{g(x)} \left\{ g(x) \cdot \frac{1}{f(x)} \cdot f'(x) + \log f(x) \cdot g'(x) \right\}$$

$$16. \frac{dy}{dx} = (-1)(2-x)(3-x) \dots (n-x) +$$

$$(-1)(1-x)(3-x) \dots (n-x) + \dots$$

$$(-1)(1-x)(2-x) \dots (n-x)$$

$$\Rightarrow \left(\frac{dy}{dx} \right)_{x=1} = (-1)(1)(2) \dots (n-1) = (-1)(n-1)!$$

$$17. \frac{dy}{dx} = \frac{\left(\frac{dy}{dt} \right)}{\left(\frac{dx}{dt} \right)} \text{ and substitute } t = \frac{1}{2}$$

$$18. \frac{dy}{dx} = \frac{\left(\frac{dy}{dt} \right)}{\left(\frac{dx}{dt} \right)}$$

$$19. \frac{dy}{dx} = \frac{\left(\frac{dy}{dt} \right)}{\left(\frac{dx}{dt} \right)}$$

$$20. \text{ Use } \frac{d^2y}{dx^2} = \frac{d\left(\frac{dy}{dx}\right)}{dt} \times \frac{1}{\left(\frac{dx}{dt}\right)} \text{ and substitute } t = \pi/2$$

$$21. \text{ Use } \frac{d^2y}{dx^2} = \frac{d\left(\frac{dy}{dx}\right)}{dt} \times \frac{1}{\left(\frac{dx}{dt}\right)} \text{ and substitute } t = 1/2$$

$$22. \text{ Put } x = \tan \theta \text{ in } \tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right)$$

$$\text{Put } x = \sin \theta$$

$$\text{in } \tan^{-1} \left(\frac{2x\sqrt{1-x^2}}{1-2x^2} \right)$$

$$23. y = e^{\sin^{-1} x}; z = \log x$$

$$\frac{dy}{dz} = \frac{\left(\frac{dy}{dx} \right)}{\left(\frac{dz}{dx} \right)}$$

$$24. \sqrt{y - \sqrt{y - \sqrt{y - \dots \infty}}} = \sqrt{x + \sqrt{x + \dots \infty}} = v$$

$$\sqrt{y-v} = \sqrt{x+v} = v$$

$$\Rightarrow \left. \begin{matrix} y = v^2 + v \\ x = v^2 - v \end{matrix} \right\} \Rightarrow \frac{y+x}{2} = v^2 \text{ and } \frac{y-x}{2} = v$$

By eliminating v we get the solution.

$$25. x = \frac{\cos y}{\cos(a+y)} \text{ and then differentiate}$$

$$26. \frac{dy}{dx} = \frac{f'(x)}{(2y-1)}$$

$$27. x\sqrt{1+y} = -y\sqrt{1+x} \Rightarrow x^2(1+y) = y^2(1+x)$$

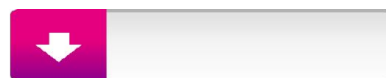
$$\Rightarrow x^2 - y^2 = xy^2 - x^2y$$

$$\Rightarrow (x-y)(x+y+xy) = 0, \text{ if } x \neq y$$

$$\Rightarrow x + y + xy = 0$$

$$\Rightarrow y(x+1) = -x \Rightarrow y = \frac{-x}{x+1}, \text{ differentiate}$$

$$28. (f \circ g)(x) = f \left(\frac{x}{\sqrt{1+x^2}} \right)$$



Formula Based:

$$1. \frac{d}{dx} \left\{ \log(x + \sqrt{a^2 + x^2}) \right\} =$$

$$1) \frac{1}{(x + \sqrt{a^2 + x^2})} \quad 2) \frac{x}{\sqrt{a^2 + x^2}}$$

$$3) \frac{1}{x(x + \sqrt{a^2 + x^2})} \quad 4) \frac{1}{\sqrt{a^2 + x^2}}$$

$$2. \text{ If } y = \sqrt{\frac{\sec x - 1}{\sec x + 1}} \text{ then } \frac{dy}{dx} =$$

- 1) $\frac{1}{2} \sec^2 \frac{x}{2}$ 2) $\sec^2 \frac{x}{2}$
 3) $\frac{1}{2} \tan \frac{x}{2}$ 4) $\tan \frac{x}{2}$
3. If $y = 5^{2\{\log_5(x+1) - \log_5(3x+1)\}}$ then $\frac{dy}{dx}$ at $x=0$ is

- 1) 0 2) $\frac{1}{3}$ 3) $\frac{3}{5}$ 4) -4

4. If $y^2 = ax^2 + bx + c$, then $y^3 \frac{d^2y}{dx^2}$ is
 1) a constant 2) a function of x only
 3) a function of y only 4) a function of x and y

5. If $e^{g(y)} - e^{-g(y)} = 2f(x)$ then $\frac{dy}{dx} =$
 1) $\frac{f'(x)}{g'(y)} \sqrt{1+f(x)^2}$ 2) $\frac{f'(x)}{g'(y)} \frac{1}{\sqrt{1+(f(x))^2}}$
 3) $\frac{f'(x)}{g'(y)} \frac{1}{\sqrt{1+(f(x))^2}}$ 4) $2 \frac{f'(x)}{g'(y)} \sqrt{1+f(x)^2}$

Inverse Trigonometric Functions and Substitutions:

6. $y = \sin^{-1} 2x + \tan^{-1} 3x + \cos^{-1} 2x + \cot^{-1} 3x$
 then $\frac{dy}{dx}$

- 1) 0 2) 1 3) $\frac{2}{\sqrt{1-x^2}} + \frac{3}{\sqrt{1+9x^2}}$ 4) 3

7. If $y = \tan^{-1} \frac{\sqrt{1+x^2} - 1}{x}$ then $\frac{dy}{dx} =$
 1) $\frac{1}{1+x^2}$ 2) $\frac{-1}{1+x^2}$ 3) $\frac{1}{2(1+x^2)}$ 4) $\frac{2}{1+x^2}$

8. $\frac{d}{dx} \left\{ \tan^{-1} \left(\frac{a \sin x + b \cos x}{a \cos x - b \sin x} \right) \right\} =$
 1) 0 2) 1 3) -1 4) $\frac{1}{1+x^2}$

9. $\frac{d}{dx} \left[\sin^2 \left(\cot^{-1} \sqrt{\frac{1+x}{1-x}} \right) \right] =$

- 1) 1 2) -1 3) $\frac{1}{2}$ 4) $\frac{-1}{2}$

10. $\frac{d}{dx} \left\{ \sin^{-1} \left(\frac{5x+12\sqrt{1-x^2}}{13} \right) \right\} =$
 1) $\frac{1}{\sqrt{1-x^2}}$ 2) $\frac{-1}{\sqrt{1+x^2}}$

- 3) $\frac{2}{\sqrt{1-x^2}}$ 4) 0

11. $\frac{d}{dx} \left\{ \cos^{-1} x + \sin^{-1} \sqrt{1-x^2} \right\} =$

- 1) 0 2) 1 3) $\frac{2}{\sqrt{1-x^2}}$ 4) $\frac{-2}{\sqrt{1-x^2}}$

Derivative of Hyperbolic and Inverse Functions:

12. $\frac{d}{dx} \left[\frac{e^{\sin \sqrt{x}} + e^{-\sin \sqrt{x}}}{e^{\sin \sqrt{x}} - e^{-\sin \sqrt{x}}} \right] =$

- 1) $\frac{-\frac{2}{\sqrt{x}} \cos \sqrt{x}}{(e^{\sin \sqrt{x}} - e^{-\sin \sqrt{x}})^2}$ 2) $\frac{-\frac{2}{\sqrt{x}} \cos \sqrt{x}}{e^{\sin \sqrt{x}}}$
 3) $\frac{-2 \cos \sqrt{x}}{e^{\sin \sqrt{x}}}$ 4) $\frac{2 \cos \sqrt{x}}{e^{\sin \sqrt{x}}}$

13. $\frac{d}{dx} \left[\tanh^{-1} \left(\frac{2x}{1+x^2} \right) \right] =$

- 1) $\frac{2}{1-x^2}$ 2) $\frac{2}{x^2-1}$
 3) $\frac{2}{1+x^2}$ 4) $\frac{-2}{x^2+1}$

Logarithmic Differentiation:

14. $\frac{d}{dx} \left\{ (x^x)^x \right\} =$

- 1) $(x^x)^x \{x(1+\log x)\}$ 2) $(x^x)^x \cdot x(1-2\log x)$
 3) $(x^x)^x (1+2\log x)x$ 4) $(x^x)^x \cdot x^2(1-2\log x)$

15. If $y = (ax+b)^{(cx+d)}$ then $\frac{dy}{dx} =$

- 1) $y \left\{ \frac{a(cx+d)}{(ax+b)} + c \log(ax+b) \right\}$
 2) $y \left\{ \frac{a(cx+d)}{(ax+b)} - c \log(ax+b) \right\}$

$$3) y \left\{ \frac{(cx+d)}{(ax+b)} + \log(ax+b) \right\}$$

$$4) y \left\{ \frac{(cx+d)}{(ax+b)} - \log(ax+b) \right\}$$

16. If $(\cos x)^y = (\sin y)^x$ then $\frac{dy}{dx} =$

1) $\frac{\log(\sin y) + y \tan x}{\log(\cos x) - x \cot y}$

2) $\frac{\log(\sin y) - y \tan x}{\log(\cos x) + \cot y}$

3) $\frac{\log(\sin y)}{\log(\cos x)}$

4) $\frac{\log(\cos x)}{\log(\sin y)}$

Parametric Differentiation:

17. If $x = \cos^{-1}\left(\frac{1}{\sqrt{1+t^2}}\right)$, $y = \sin^{-1}\left(\frac{t}{\sqrt{1+t^2}}\right)$

Then $\frac{dy}{dx} =$

1) 0 2) 1 3) -1 4) 2

18. $x = a \sin 2\theta (1 + \cos 2\theta)$,

$y = b \cos 2\theta (1 - \cos 2\theta)$, then $\frac{dy}{dx} =$

1) $\frac{a}{b} \tan \theta$

2) $\frac{b}{a} \tan \theta$

3) $\frac{a}{b} \cot \theta$

4) $\frac{b}{a} \cot \theta$

19. If $x^2 + y^2 = t + \frac{1}{t}$ and $x^4 + y^4 = t^2 + \frac{1}{t^2}$

then $x^3 y \frac{dy}{dx} =$ [EAM-2012]

1) 0 2) 1 3) -1 4) 2

20. If $x = \cos^3 \theta$, $y = \sin^3 \theta$ then $\sqrt{1 + \left(\frac{dy}{dx}\right)^2} =$

1) $\tan^2 \theta$

2) $\sec^2 \theta$

3) $\sec \theta$

4) $|\sec \theta|$

Derivative of one function w.r.t. another function:

21. The derivative of $\sec^{-1}\left(\frac{1}{2x^2-1}\right)$ with respect to $\sqrt{1+3x}$ at $x = -\frac{1}{3}$

1) 0

2) $\frac{1}{2}$

3) $\frac{1}{4}$

4) $\frac{1}{8}$

Derivative of implicit function:

22. If $y = \sqrt{x + \sqrt{y + \sqrt{x + \sqrt{y + \dots \infty}}}}$ then $\frac{dy}{dx}$ is equal to

1) $\frac{y^2 + x}{2y^3 - 2xy - 1}$

2) $\frac{y^2 - x}{2y^3 - 2xy - 1}$

3) $\frac{y^2 - x}{2y^3 - 2xy + 1}$

4) $\frac{y^2}{2y^3 - 2xy + 1}$

23. If $x \sin y = 3 \sin y + 4 \cos y$, then $\frac{dy}{dx} =$

1) $\frac{-\sin^2 y}{4}$

2) $\frac{\sin^2 y}{4}$

3) $\frac{-\cos^2 y}{4}$

4) $\frac{\cos^2 y}{4}$

24. If $y = \sqrt{\log(\cos 2x) + y}$ then $\frac{dy}{dx} =$

1) $\frac{2 \tan 2x}{(2y-1)}$

2) $\frac{\tan 2x}{(2y-1)}$

3) $\frac{-2 \tan 2x}{(2y-1)}$

4) $\frac{1}{(2y-1)}$

25. If $y = \frac{1}{x}$ then $\frac{dy}{\sqrt{1+y^4}} + \frac{dx}{\sqrt{1+x^4}} =$

1) 0

2) 1

3) $\frac{x}{y}$

4) $\frac{y}{x}$

Derivative of composite function:

26. If g is the inverse of f and $f'(x) = \frac{1}{2+x^n}$, then

$g'(x)$ is equal to

- 1) $2+x^n$ 2) $2+(f(x))^n$
 3) $2+(g(x))^n$ 4) 0

LEVEL-II (H.W)-KEY

- 1) 4 2) 1 3) 4 4) 1 5) 2
 6) 1 7) 3 8) 2 9) 4 10) 1
 11) 4 12) 1 13) 1 14) 3 15) 1
 16) 1 17) 2 18) 2 19) 3 20) 4
 21) 1 22) 2 23) 1 24) 3 25) 1
 26) 3

LEVEL-II (H.W)-HINTS

1. $\frac{d}{dx} \left\{ \log \left(x + \sqrt{a^2 + x^2} \right) \right\}$

$$= \frac{d}{dx} \left(\frac{\sinh^{-1} \frac{x}{a}}{1} \right) = \frac{1}{\sqrt{x^2 + a^2}}$$

2. $y = \sqrt{\frac{2 \sin^2 \frac{x}{2}}{2 \cos^2 \frac{x}{2}}} = \tan \frac{x}{2}$

3. $y = 5^{2 \log_5 \left(\frac{x+1}{3x+1} \right)} = 5^{\log_5 \left(\frac{x+1}{3x+1} \right)^2} = \left(\frac{x+1}{3x+1} \right)^2$

4. $y^2 = ax^2 + bx + c$

$$\Rightarrow 2y \frac{dy}{dx} = 2ax + b \Rightarrow 2 \left(\frac{dy}{dx} \right)^2 + 2y \frac{d^2y}{dx^2} = 2a$$

$$\Rightarrow y \frac{d^2y}{dx^2} = a - \left(\frac{dy}{dx} \right)^2$$

$$\Rightarrow y \frac{d^2y}{dx^2} = a - \left(\frac{2ax + b}{2y} \right)^2$$

LEVEL - III

$$\Rightarrow y \frac{d^2y}{dx^2} = \frac{4ay^2 - (2ax + b)^2}{4y^2}$$

$$4y^3 \frac{d^2y}{dx^2} = 4ac - b^2 \text{ is a constant}$$

5. $e^{g(y)} g'(y) \frac{dy}{dx} + e^{-g(y)} g'(y) \frac{dy}{dx} = 2f'(x)$

$$\frac{dy}{dx} = \frac{f'(x)}{g'(y)} \cdot \frac{1}{\sqrt{1 + (f'(x))^2}}$$

6. $y = (\sin^{-1} 2x + \cos^{-1} 2x) + (\tan^{-1} 3x + \cot^{-1} 3x)$

$$y = \frac{\pi}{2} + \frac{\pi}{2} = \pi \Rightarrow \frac{dy}{dx} = 0$$

7. Put $x = \tan \theta$

8. Put $a = \cos \alpha, b = \sin \alpha$, then

$$\tan^{-1} \left(\frac{\sin(x + \alpha)}{\cos(x + \alpha)} \right) = \tan^{-1} (\tan(x + \alpha))$$

9. Put $x = \cos \theta$

10. $\frac{d}{dx} \left\{ \sin^{-1} \left(\frac{5}{13}x + \frac{12}{13}\sqrt{1-x^2} \right) \right\},$

$$x = \sin \theta, \cos \alpha = \frac{5}{13}, \sin \alpha = \frac{12}{13}$$

11. Put $x = \cos \theta$

12. Given $= \frac{d}{dx} \left\{ \coth(\sin \sqrt{x}) \right\}$

13. $\frac{d}{dx} (\tanh^{-1} x) = \frac{1}{1-x^2}$

14. $y = (x^x)^x \Rightarrow \log y = x \log(x^x) = x^2 \log x$
 then differentiate w.r.t.x

15. $\frac{d}{dx} \left\{ f(x)^{g(x)} \right\} =$

$$f(x)^{g(x)} \left\{ g(x) \cdot \frac{1}{f(x)} \cdot f'(x) + \log f(x) \cdot g'(x) \right\}$$

16. Take logarithm on both sides

17. Put $t = \tan \theta$

18. $\frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta} \right)}{\left(\frac{dx}{d\theta} \right)}$

19. $(x^2 + y^2)^2 = \left(t + \frac{1}{t} \right)^2$

$$\Rightarrow x^4 + y^4 + 2x^2y^2 = t^2 + \frac{1}{t^2} + 2$$

$$\Rightarrow x^4 + y^4 + 2x^2y^2 = x^4 + y^4 + 2$$

$$\Rightarrow x^2y^2 = 1 \text{ and then differentiate}$$

$$20. \quad \frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)}$$

$$21. \quad y = \sec^{-1}\left(\frac{1}{2x^2-1}\right), \quad z = \sqrt{1+3x}$$

$$(x = \cos \theta) \quad \frac{dy}{dz} = \frac{\left(\frac{dy}{dx}\right)}{\left(\frac{dz}{dx}\right)}$$

$$22. \quad y = \sqrt{x + \sqrt{y+y}} \Rightarrow y^2 = x + \sqrt{2y}$$

$$23. \quad \sin y(x-3) = 4 \cos y \text{ and then differentiate}$$

$$24. \quad \text{Squaring on both sides and then differentiate.}$$

$$25. \quad \frac{dy}{dx} = -\frac{1}{x^2}$$

$$\frac{dy}{dx} + \frac{\sqrt{1+y^4}}{\sqrt{1+x^4}} = \frac{dy}{dx} + \frac{1}{x^2} = -\frac{1}{x^2} + \frac{1}{x^2} = 0$$

$$26. \quad g(x) = f^{-1}(x) \Rightarrow (f \circ g)(x) = x \Rightarrow f[g(x)]g'(x) = 1$$

$$\Rightarrow f'[g(x)] = \frac{1}{g'(x)} \Rightarrow \frac{1}{2+[g(x)]^n} = \frac{1}{g'(x)}$$

$$\text{then } g'(x) = 2 + [g(x)]^n$$



$$1. \quad \text{If } (f(x))^{g(y)} = e^{f(x)-g(y)} \text{ then } \frac{dy}{dx} =$$

$$1) \frac{f'(x) \log f(x)}{g(y)(1+\log f(x))^2} \quad 2) \frac{f'(x) \log f(x)}{g'(y)(1+\log f(x))^2}$$

$$3) \frac{f(x) \cdot \log f(x)}{g'(y)(1-\log f(x))^2} \quad 4) 2 \frac{f'(x) \log f(x)}{g(y)(1+\log f(x))^2}$$

$$2. \quad \text{If } \cos^{-1}\left(\frac{x^2 - y^2}{x^2 + y^2}\right) = \log a \text{ then } \frac{dy}{dx} =$$

$$1) -\frac{x}{y} \quad 2) -\frac{y}{x} \quad 3) \frac{y}{x} \quad 4) \frac{x}{y}$$

$$3. \quad \text{If } f(x) = (x-a)g(x) \text{ and } g(x) \text{ is continuous at } x=a \text{ then } f'(1) =$$

$$1) a g(1) \quad 2) g'(1) \quad 3) g(1) \quad 4) a g'(1)$$

$$4. \quad \text{Let } f: \mathbb{R} \rightarrow \mathbb{R} \text{ such that for all 'x' and y in } \mathbb{R}, |f(x) - f(y)| \leq |x - y|^3 \text{ then } f(x)$$

$$1) e^x \quad 2) e^{-x} \quad 3) x \quad 4) 'c' (\text{constant})$$

$$5. \quad \phi(x) = f(x)g(x) \text{ and } f'(x)g'(x) = k,$$

$$\text{then } \frac{2k}{f(x)g(x)} =$$

$$1) \frac{\phi''(x)}{\phi(x)} - \frac{f''(x)}{f(x)} - \frac{g''(x)}{g(x)}$$

$$2) \frac{\phi''(x)}{\phi(x)} + \frac{f''(x)}{f(x)} + \frac{g''(x)}{g(x)}$$

$$3) \frac{\phi''(x)}{\phi(x)} + \frac{f''(x)}{f(x)} - \frac{g''(x)}{g(x)}$$

$$4) \frac{\phi''(x)}{\phi(x)} - \frac{f''(x)}{f(x)} + \frac{g''(x)}{g(x)}$$

$$6. \quad \text{If } f(x) = \frac{g(x)+g(-x)}{2} + \frac{2}{[h(x)+h(-x)]^{-1}} \text{ where } g \text{ and } h \text{ are differentiable functions}$$

$$\text{then } f'(0) =$$

$$1) 1 \quad 2) 0 \quad 3) \frac{1}{2} \quad 4) \frac{3}{2}$$

$$7. \quad \text{If } x = \sqrt{\frac{1-t^2}{1+t^2}}, y = \frac{\sqrt{1+t^2} - \sqrt{1-t^2}}{\sqrt{1+t^2} + \sqrt{1-t^2}} \text{ then}$$

$$\frac{dy}{dx} =$$

$$1) \frac{1-x}{1+x} \quad 2) \frac{-2}{(1+x)^2} \quad 3) \frac{2}{(1+x)^2} \quad 4) \frac{1+x}{1-x}$$

$$8. \quad \frac{d}{dx} \left\{ e^{\frac{1}{2} \log(1-\tan h^2 x)} + 3^{\frac{1}{2} \log_3(\cot h^2 x - 1)} \right\} =$$

- 1) $\frac{-(\sin h^3 x + \cos h^3 x)}{\cos h^2 x \sin h^2 x}$ 2) $\frac{\sin h^3 x + \cos h^3 x}{\cos h^2 x \sin h^2 x}$
- 3) $\frac{\sin h^2 x + \cos h^2 x}{\sin h^2 x \cos h^2 x}$ 4) $\sec h^2 x \cos h^2 x$
9. If $x < 1$, then
- $$\frac{1}{1+x} + \frac{2x}{1+x^2} + \frac{4x^3}{1+x^4} + \dots \infty =$$
- 1) x 2) $\frac{1}{1+x}$ 3) $\frac{1}{1-x}$ 4) $\frac{1}{x}$
10. $f(x) = [\tan \pi/4 + \tan x][\tan \pi/4 + \tan(\pi/4 - x)]$ and $g(x) = x(x+1)$. then the value of $g^1[f(x)]$ is equal to
- 1) $\frac{1}{1+x^2}$ 2) 4 3) 5 4) $\frac{2x}{1+x^2}$
11. If $\sqrt{\tan y} = e^{\cos 2x} \cdot \sin x$, then $\frac{dy}{dx} =$
- 1) $\sin 2y \cdot (\cot x - 2 \sin 2x)$
 2) $\sin 2x (\cot y - \sin y)$
 3) $\sin 2y \cdot \sin 2x$ 4) $\cos 2y \cdot \cos 2x$
12. If $x = \cos \theta$, $y = \sin 5\theta$ then
- $$(1-x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} =$$
- 1) $-5y$ 2) $5y$ 3) $25y$ 4) $-25y$
13. If $P(x) = ax^3 + bx^2 + cx + d$ and $p(0) = 4$, $P^1(0) = 3$, $P^{11}(0) = 4$, $P^{111}(0) = 6$ then arrange the values of a, b, c, d in the descending order of their values
- 1) a, b, c, d 2) a, b, d, c
 3) d, c, b, a 4) b, a, c, d
14. Let $f(x) = x^n$, n being a positive integer. The value of n for which $f^1(a+b) = f^1(a) + f^1(b)$, when $a, b > 0$ is
- 1) 1 2) 2 3) 3 4) 4
15. If $f(x) = |x-2|$, $g(x) = f(f(x))$, then for $x > 4$, $g^1(x) =$
- 1) 0 2) 1 3) -1 4) 2
16. If $\frac{d}{dx} \left(\frac{1+x^2+x^4}{1+x+x^2} \right) = ax + b$, then $(a, b) =$
- 1) $(-1, 2)$ 2) $(-2, 1)$
 3) $(2, -1)$ 4) $(1, 2)$

17. If $y = \log \cos \left(\tan^{-1} \left(\frac{e^x - e^{-x}}{2} \right) \right)$, then $\frac{dy}{dx} =$
- 1) $-\tanh x$ 2) $\sinh x$
 3) $\cosh x$ 4) $\coth x$
18. $\frac{d}{dx} \left[\log_e \left\{ (e^x + 2) + \sqrt{e^{2x} + 4e^x + 5} \right\} \right] =$
- 1) $\frac{1}{\sqrt{e^{2x} + 4e^x + 5}}$ 2) $\frac{e^x}{\sqrt{e^{2x} + 4e^x + 5}}$
 3) $\frac{e^x}{\sqrt{e^{2x} + 4e^x + 3}}$ 4) $\frac{-e^x}{\sqrt{e^{2x} + 4e^x + 3}}$
19. If $f(x) = (ax + b) \cos x + (cx + d) \sin x$ and $f^1(x) = x \cos x$ is an identity in x then a, b, c, d are given by
- 1) $a = b = c = d$ 2) 0, 1, -1, 0
 3) 1, 0, -1, 0 4) 0, 1, 1, 0
20. If $f(x) = \begin{vmatrix} \sin x & 1 & 0 \\ x - \frac{\pi}{2} & \sin x & 1 \\ 0 & 1 & \sin x \end{vmatrix}$ then
- $$\frac{df}{dx} \text{ at } x = \frac{\pi}{2} \text{ is}$$
- 1) 2 2) $\frac{\pi}{2}$ 3) 1 4) 8
21. If $y = \log \left\{ \left(\frac{1+x}{1-x} \right)^{\frac{1}{4}} \right\} - \frac{1}{2} \tan^{-1}(x)$, then
- $$\frac{dy}{dx} =$$
- 1) $\frac{x}{1-x^2}$ 2) $\frac{x^2}{1-x^4}$ 3) $\frac{x}{1+x^4}$ 4) $\frac{x}{1-x^4}$
22. If $f^1(x) = g(x)$ and $g^1(x) = -f(x)$ for all x and $f(2) = 4 = f^1(2)$ then $f^2(24) + g^2(24)$ is
- 1) 32 2) 24 3) 64 4) 48
23. If $f(x) = (\cos x + i \sin x)(\cos 3x + i \sin 3x) \dots (\cos(2n-1)x + i \sin(2n-1)x)$, then $f^{11}(x)$ is
- 1) $n^2 f(x)$ 2) $-n^4 f(x)$ 3) $-n^2 f(x)$ 4) $n^4 f(x)$

24. If $\sqrt{x+y} + \sqrt{y-x} = c$, then $\frac{d^2y}{dx^2} =$
 1) $\frac{2}{c}$ 2) $\frac{-2}{c^2}$ 3) $\frac{2}{c^2}$ 4) $\frac{-2}{c}$
25. Let $y=e^{2x}$. Then $\left(\frac{d^2y}{dx^2}\right)\left(\frac{d^2x}{dy^2}\right) =$
 1) 1 2) e^{-2x} 3) $2e^{-2x}$ 4) $-2e^{-2x}$
26. If $f(x) = \sqrt{x^2 - 2x + 1}$, then
 1) $f'(x) = 1, \forall x$ 2) $f'(x) = 1, \forall x \geq 1$
 3) $f'(x) = -1, \forall x \leq 1$
 4) $f'(x) = 1$ if $x > 1$ and $f'(x) = -1$ if $x < 1$
27. If $(\sin x)(\cos y) = 1/2$ then $\frac{d^2y}{dx^2}$ at $\left(\frac{\pi}{4}, \frac{\pi}{4}\right)$ is equal to
 1) -4 2) -2 3) -6 4) 0
28. If $f(x) = \operatorname{sgn}(\cos x)$, then $f'\left(\frac{\pi}{2}\right)$ is
 1) 0 2) -1 3) 1 4) 2
29. If $y^2 = p(x)$, a polynomial of degree 3, then $2 \frac{d}{dx} \left(y^3 \frac{d^2y}{dx^2} \right)$ is equal to
 1) $p^{111}(x) + p^1(x)$ 2) $p^{11}(x) + p^{111}(x)$
 3) $p(x)p^{111}(x)$ 4) constant
30. If $y = \cos^{-1} \frac{9-x^2}{9+x^2}$ then $y'(-1)$ is equal to
 1) -3/5 2) 3/5 3) 2/7 4) 3/8
31. Let a function $y=y(x)$ be defined parametrically by $x=2t-|t|, y=t^2+t|t|$. Then $y'(x), x>0$
 1) 0 2) $4x$ 3) $2x$ 4) x
32. If $t = \frac{2\sqrt{2} - (1+\sqrt{3})}{\sqrt{3}-1}$ and $f(x) = \frac{2x}{1-x^2}, g(x) = \frac{3x-x^3}{1-3x^2}$ then $\frac{d}{dt} \{f(g(t))\} =$
 1) 1 2) 2 3) 0 4) -1
33. If $f(x) = \cos x \cos 2x \cos 4x \cos 8x \cos 16x$ then $f'\left(\frac{\pi}{4}\right)$ equals
 1) $-\sqrt{2}$ 2) 0
 3) $\frac{1}{\sqrt{2}}$ 4) $\operatorname{Cosec} \frac{\pi}{4}$
34. If $f(x) = |x-8|$ & $g(x) = f(f(x)) \forall x > 16$ then $g'(x)$

equals

- 1) 1 2) -1 3) 0 4) 2

35. If $y^{y^y} = \log_e(x + \log_e(x + \dots))$ then $\frac{dy}{dx}$ at

 $(x=e^2-2, y=\sqrt{2})$ is

- 1) $\frac{\log(e/2)}{2\sqrt{2}(e^2-1)}$ 2) $\frac{\log 2}{2\sqrt{2}(e^2-1)}$
 3) $\frac{\sqrt{2}\log(e/2)}{(e^2-1)}$ 4) $\frac{\log(e/2)}{(e^2-1)}$

36. If $y = \frac{\sin^2 x}{1+\cot x} + \frac{\cos^2 x}{1+\tan x}$ then $y'(x)$ is equal to

- 1) $\cos 2x$ 2) $-\cos 2x$
 3) $2\cos^2 x$ 4) $\cos^3 x$

37. Let f be a function defined for every x , such that $f^{11} = -f, f(0)=0, f'(0)=1$ then $f(x)$ is equal to

- 1)
- $\tan x$
- 2)
- $e^x - 1$
- 3)
- $\sin x$
- 4)
- $2\sin x$

38. If $t = \sin^{-1} 2^s$ then $\frac{ds}{dt}$ is equal to

- 1) $\frac{\log 2}{\sqrt{1-t^2}}$ 2) $\frac{\sin t}{\log 2}$ 3) $\frac{\cot t}{\log 2}$ 4) $\frac{\tan t}{\log 2}$

39. If $y = \frac{x}{a + \frac{x}{b + \frac{x}{a + \frac{x}{b + \dots + to \infty}}}}$ then $\frac{dy}{dx} =$

- 1) $\frac{1}{a(2y+b)}$ 2) $\frac{b}{a(2y+b)}$
 3) $\frac{1}{ab(2y+b)}$ 4) $\frac{ab}{(2y+b)}$

40. If $f(x) = |x|^{\sin x}$, then $f'\left(-\frac{\pi}{4}\right)$ is equals

- 1) $\left(\frac{\pi}{4}\right)^{\sqrt{2}} \left(\frac{\sqrt{2}}{2} \ln \frac{4}{\pi} - \frac{2\sqrt{2}}{\pi} \right)$
 2) $\left(\frac{\pi}{4}\right)^{\sqrt{2}} \left(\frac{\sqrt{2}}{2} \ln \frac{4}{\pi} + \frac{2\sqrt{2}}{\pi} \right)$
 3) $\left(\frac{\pi}{4}\right)^{\sqrt{2}} \left(\frac{\sqrt{2}}{2} \ln \frac{\pi}{4} - \frac{2\sqrt{2}}{\pi} \right)$

- 4) $\left(\frac{\pi}{4}\right)^{\sqrt{2}} \left(\frac{\sqrt{2}}{2} \ln \frac{\pi}{4} + \frac{2\sqrt{2}}{\pi}\right)$
41. If $Y = \tan^{-1}\left(\frac{1}{1+x+x^2}\right) + \tan^{-1}\left(\frac{1}{x^2+3x+3}\right) + \tan^{-1}\left(\frac{1}{x^2+5x+7}\right) + \dots$ upto n terms then $y'(0) =$
- 1) $\frac{-1}{1+n^2}$ 2) $\frac{-n^2}{1+n^2}$ 3) $\frac{n}{1+n^2}$ 4) n
42. If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$ then $\frac{dy}{dx} =$
- 1) $\sqrt{\frac{1-x^2}{1-y^2}}$ 2) $\sqrt{(1-x^2)(1-y^2)}$
- 3) $\sqrt{\frac{1-y^2}{1-x^2}}$ 4) $\frac{1}{\sqrt{(1-x^2)(1-y^2)}}$
43. If $f(x) = 3e^{x^2}$, then $f'(x) - 2xf(x) + \frac{1}{3}f(0) - f'(0)$ is equal to
- 1) 0 2) 1 3) $\frac{7}{3}e^{x^2}$ 4) e^{x^2}
44. If $f'(x) = \sin(\log x)$ and $y = f\left(\frac{2x+3}{3-2x}\right)$, then $\frac{dy}{dx}$ is
- 1) $\sin(\log x) \cdot \frac{1}{x \log x}$
- 2) $\frac{12}{(3-2x)^2} \sin\left[\log\left(\frac{2x+3}{3-2x}\right)\right]$
- 3) $\sin\left[\log\left(\frac{2x+3}{3-2x}\right)\right]$ 4) $\sin(\log x)$
45. If $f(x) = \log_x(\log x)$, then $f'(x)$ at $x=e$ is
- 1) 0 2) 1 3) $\frac{1}{e}$ 4) $\frac{1}{2e}$
46. $y = \log^n x$ where $\log^n$ means $\log \cdot \log \dots$ (repeated n times) then $x \cdot \log x \cdot \log^2 x \cdot \log^3 x \dots - \log^{n-1} x \cdot \frac{dy}{dx} =$
- 1) $\log x$ 2) $\log^n x$ 3) $\frac{1}{\log x}$ 4) 1

47. $\frac{d^2x}{dy^2} =$ [AIEEE - 2011]

- 1) $\frac{1}{\left(\frac{dy}{dx}\right)^2}$ 2) $\frac{\left(\frac{d^2y}{dx^2}\right)}{\left(\frac{dy}{dx}\right)^2}$
- 3) $\left(\frac{d^2y}{dx^2}\right)$ 4) $\frac{-\left(\frac{d^2y}{dx^2}\right)}{\left(\frac{dy}{dx}\right)^3}$

48. If $y = \left(x + \sqrt{x^2 - a^2}\right)^n$ then $(x^2 - a^2) \left(\frac{dy}{dx}\right)^2 =$

1) n^2y 2) $-n^2y$ 3) ny^2 4) n^2y^2

49. If $|x| < 1$, then

$$\frac{d}{dx} \left[1 + \frac{p}{q}x + \frac{P(p+q)}{\angle 2} \left(\frac{x}{q}\right)^2 + \frac{p(p+q)(p+2q)}{\angle 3} \left(\frac{x}{q}\right)^3 + \dots \right] =$$

- 1) $\frac{p}{q(1-x)^{\frac{p}{q}+1}}$ 2) $\frac{p}{q(1-x)^{\frac{p}{q}}}$
- 3) $(1-x)^{-pq-1}$ 4) $(1-x)^{pq+1}$

50. $\frac{d}{dx} \left[\frac{\tan x - \cot x}{\tan x + \cot x} \right] =$

- 1) $2 \sin 2x$ 2) $-2 \sin 2x$
- 3) $2 \cos 2x$ 4) $-2 \cos 2x$

51. If $y = \cot^{-1}(1+x^2-x)$, then $\frac{dy}{dx} =$

- 1) $\frac{1}{1+x^2} + \frac{1}{1+(x-1)^2}$ 2) $\frac{1}{1+(x-1)^2} - \frac{1}{1+x^2}$
- 3) $\frac{1}{1+x^2} - \frac{1}{1+(x-1)^2}$ 4) $-\frac{1}{1+x^2} - \frac{1}{1+(x-1)^2}$

52. $\frac{d}{dx} (\sqrt{\sec^{-1} x^2}) =$

- 1) $\frac{1}{x\sqrt{\sec^{-1} x^2} \cdot \sqrt{x^4-1}}$ 2) $\frac{x}{\sqrt{\sec^{-1} x^2} \cdot \sqrt{x^4-1}}$
- 3) $\frac{-1}{x\sqrt{\sec^{-1} x^2} \cdot \sqrt{x^4-1}}$ 4) $\frac{-1}{\sqrt{\sec^{-1} x^2} \cdot \sqrt{x^4-1}}$

53. $\frac{d}{dx} \left\{ \frac{1}{2} \cot h \frac{x}{2} - \frac{1}{6} \cot h^3 \frac{x}{2} \right\} =$

- 1) $-\frac{1}{4} \operatorname{cosec} h \frac{4x}{2}$ 2) $-\frac{1}{4} \cot h^4 x / 2$
 3) $\frac{1}{4} \operatorname{cosec} h^4 x / 2$ 4) $\frac{1}{4} \cot h^4 x / 2$
54. If $x = \sec \theta - \cos \theta$ and $y = \sec^n \theta - \cos^n \theta$ then $\left(\frac{dy}{dx}\right)^2$ is equal to
 1) $\frac{n^2(y^2+4)}{x^2+4}$ 2) $\frac{n^2(y^2-4)}{x^2}$
 3) $n \frac{(y^2-4)}{x^2-4}$ 4) $\left(\frac{ny}{x}\right)^2 - 4$
55. If $y = \sin x \left[\frac{1}{\sin x \cdot \sin 2x} + \frac{1}{\sin 2x \cdot \sin 3x} + \dots + \frac{1}{\sin nx \cdot \sin(n+1)x} \right]$ then $\frac{dy}{dx} =$
 1) $\cot x - \cot(n+1)x$
 2) $(n+1) \operatorname{cosec}^2(n+1)x - \operatorname{cosec}^2 x$
 3) $\operatorname{cosec}^2 x - (n+1) \operatorname{cosec}^2(n+1)x$
 4) $\cot x + \cot(n+1)x$
56. The value of $3f(x) - 2f\left(\frac{1}{x}\right) = x$ then $f'(2) =$
 1) $\frac{2}{7}$ 2) $\frac{1}{2}$ 3) 2 4) $\frac{7}{2}$
57. If $x = \exp \left\{ \tan^{-1} \left(\frac{y-x^2}{x^2} \right) \right\}$ then $\frac{dy}{dx} =$
 1) $2x[1 + \tan(\log x)] + x \cdot \sec^2(\log x)$
 2) $x[1 + \tan(\log x)] + \sec^2(\log x)$
 3) $2x[1 + \tan(\log x)] + x^2 \sec^2(\log x)$
 4) $2x[1 + \tan(\log x)] + \sec^2(\log x)$
58. If $af(\tan x) + bf(\cot x) = x$ then $f'(\cot x) =$
 1) $\frac{1}{a-b}$ 2) $\frac{\sin^2 x}{a+b}$
- 3) $\frac{\sin^2 x}{a-b}$ 4) $\frac{\sin^2 x}{b-a}$
59. If $x = a \sec^2 \theta, y = a \tan^3 \theta$ then $\frac{d^3 y}{dx^3} =$
 1) $\frac{-3}{8a^2} \cot^3 \theta$ 2) $\frac{3}{8a^2} \cot^3 \theta$
 3) $3 \sec^2 \theta \tan \theta$ 4) $\frac{3}{4a} \cot \theta$
60. Let $f(x) = x^5 + 2x^3 + 3x + 4$ then the value of $28 \frac{d}{dx} (f^{-1}(x))$ at $x = -2$ is
 1) 1 2) 2 3) $1/14$ 4) -2
61. If $f(x) = \cos x \cos 2x \cos 2^2 x \cos 2^3 x \dots \cos 2^{n-1} x$ and $n > 1$, then $f'\left(\frac{\pi}{2}\right)$ is
 1) 1 2) 0
 3) -1 4) $(-1)^{n-1}$
62. If g is the inverse function of f and $f'(x) = \sin x$ then $g'(x)$ is
 1) $\operatorname{cosec}(g(x))$ 2) $\sin(g(x))$
 3) $-\frac{1}{\sin(g(x))}$ 4) $\cos(g(x))$
63. If $f(1) = 3, f'(1) = -\frac{1}{3}$, then the derivative of $\{x^{11} + f(x)\}^{-2}$ at $x = 1$, is
 1) $-\frac{1}{2}$ 2) -1 3) 1 4) $f'(1)$
64. If the function $f(x) = x^3 + e^{\frac{x}{2}}$ and $g(x) = f^{-1}(x)$, then the value of $g'(1)$ is
 1) $\frac{1}{2}$ 2) 2 3) 1 4) $-\frac{1}{2}$
65. If $\sin y + e^{-x \cos y} = e$ then $\frac{dy}{dx}$ at $(1, \pi)$ is

equal to

- 1) $\sin y$ 2) $-x \cos y$
 3) e 4) $\sin y - x \cos y$

66. If $f(x) = \begin{vmatrix} \sec \theta & \tan^2 \theta & 1 \\ \theta \sec x & \tan x & x \\ 1 & \tan x - \tan \theta & 0 \end{vmatrix}$ then $f'(\theta)$ is

- 1) 0 2) -1
 3) independent of θ 4) none

67. If $u = f(x^2)$, $v = g(x^3)$, $f'(x) = \sin x$, $g'(x) = \cos x$ then find $\frac{du}{dv}$

- 1) 1 2) $\frac{2}{3}$ 3) $\frac{2 \sin x^2}{3x \cdot \cos x^3}$ 4) $\frac{2x^2}{3x^3}$

68. If $f(0) = 0$, $f'(0) = 2$, then the derivative of $y = f(f(f(f(x))))$ at $x = 0$ in

- 1) 2 2) 8 3) 16 4) 4

69. If $x = \phi(t)$, $y = \psi(t)$ then $\frac{d^2 y}{dx^2}$ is

- 1) $\frac{\phi^1 \psi^{11} - \psi^1 \phi^{11}}{(\phi^1)^2}$ 2) $\frac{\phi^1 \psi^{11} - \psi^1 \phi^{11}}{(\phi^1)^3}$
 3) $\frac{\phi^{11}}{\psi^{11}}$ 4) $\frac{\psi^{11}}{\phi^{11}}$

70. If $u = x^2 + y^2$, and $x = s + 3t$, $y = 2s - t$, then $\frac{d^2 u}{ds^2}$ is equal to

- 1) 12 2) 32 3) 36 4) 10

71. Let $y = t^{10} + 1$, and $x = t^8 + 1$, then $\frac{d^2 y}{dx^2}$ is

- 1) $\frac{5}{2}t$ 2) $20t^8$ 3) $\frac{5}{16t^6}$ 4) $\frac{15}{16t^6}$

72. $\frac{d}{dx} \left[\cos^2 \left(\tan^{-1} \left(\sin \left(\cot^{-1} x \right) \right) \right) \right] =$

- 1) $\frac{2}{(x^2 + 2)^2}$ 2) $\frac{2x}{(x^2 + 2)^2}$
 3) $\frac{x^2 + 1}{x^2 + 2}$ 4) $\frac{-2x}{(x^2 - 1)^2}$

73. If $x \sin y = \sin(y + a)$ and

$\frac{dy}{dx} = \frac{A}{1 + x^2 - 2x \cos a}$ then the value of A is

- 1) 2 2) $\cos a$ 3) $\sin a$ 4) -2

74. If $y = \tan^{-1} \left(\frac{1}{\cos^2 x + \cos x + 1} \right) +$

$\tan^{-1} \left(\frac{1}{\cos^2 x + 3 \cos x + 3} \right) + \dots$ upto n terms then $f'(0) =$

- 1) 0 2) 1 3) $\frac{\pi}{2}$ 4) π

75. $f(x) = \cot^{-1} \left(\frac{x^x - x^{-x}}{2} \right)$ then $f'(1) =$

- 1) $-\log 2$ 2) $\log 2$ 3) 1 4) -1

76. If $y = \left(1 + \frac{1}{x} \right)^x$ then $\frac{2\sqrt{y_2(2) + \frac{1}{8}}}{\log \frac{3}{2} - \frac{1}{3}}$ is

- 1) 1 2) 3 3) 0 4) $1/3$

77. If $y = \frac{\sin x}{1 + \frac{\cos x}{1 + \frac{\sin x}{1 + \cos x \dots \infty}}}$ then $\frac{dy}{dx} =$

- 1) $\frac{(1+y)\cos x + y \sin x}{1 + 2y + \cos x - \sin x}$
 2) $\frac{(1+y)\sin x + y \cos x}{1 + 2y + \cos x - \sin x}$
 3) $\frac{(1+y)\cos x - y \sin x}{1 + 2y - \cos x + \sin x}$
 4) $\frac{(1+y)\cos x + y \sin x}{1 + 2y - \cos x - \sin x}$

78. If $\sqrt{\frac{v}{\mu}} + \sqrt{\frac{\mu}{v}} = 6$, then $\frac{dv}{d\mu} =$

- 1) $\frac{17\mu - v}{\mu - 17v}$ 2) $\frac{\mu - 17v}{17\mu - v}$

- 3) $\frac{17\mu + \nu}{\mu - 17\nu}$ 4) $\frac{\mu + 17\nu}{17\mu - \nu}$
79. If $y = \tan^{-1}\left(\frac{4\sin 2x}{\cos 2x - 6\sin^2 x}\right)$ then $\frac{dy}{dx}$ at $x = 0$ is
1) 10 2) 12 3) 6 4) 8
80. $y = \tan^{-1}\left(\frac{\log\left(\frac{e}{x^2}\right)}{\log(ex^2)}\right) + \tan^{-1}\left(\frac{3+2\log x}{1-6\log x}\right)$
then $\frac{d^2y}{dx^2} =$
1) 2 2) 1 3) 0 4) -1
81. $y = \cot^{-1}\left[\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}}\right]$ then $\frac{dy}{dx} =$
1) $1/2$ 2) $2/3$ 3) 3 4) 1
82. If $x^2 + y^2 = 2$ and $y^{11} + A \cdot y^{-3} = 0$ then $A =$
1) 0 2) 2 3) 1 4) 3
83. If $p(x)$ is a polynomial such that $p(x^2 + 1) = \{p(x)\}^2 + 1$ and $p(0) = 0$ then $p'(0)$ is equal to
1) -1 2) 0 3) 1 4) 2
84. $\frac{d}{dx}\left\{e^{x^e} + x^{e^x} + e^{x^x}\right\} =$
1) $e^{x^e} + x^{e^x} + e^{x^x}$
2) $x^2 \cdot e^{x^e} + e^x \cdot x^{e^x} + x^x \cdot e^{x^x}$
3) $ee^{x^e} \cdot x^{e-1} + x^{e^x} \cdot e^x \left(\frac{1}{x} + \log x\right) + e^{x^x} \cdot x^x (1 + \log x)$
4) $ee^{x^e} \cdot x^{e-1} + x^{e^x} \cdot e^x \left(\frac{1}{x} - \log x\right) + e^{x^x} \cdot x^x (1 - \log x)$
85. If $\sin \theta \sin(2\alpha + \theta) \sin(4\alpha + \theta) \dots$
 $\sin(2(n-1)\alpha + \theta) = \frac{\sin n\theta}{2^{n-1}}$ where $2n\alpha = \pi$ then

- $\cot(\theta) + \cot(2\alpha + \theta) + \cot(4\alpha + \theta) + \dots$
 $+ \cot(2(n-1)\alpha + \theta) =$
1) $-n \cot n\theta$ 2) $n \cot n\theta$
3) $n \tan n\theta$ 4) $-n \tan n\theta$
86. The derivative of $y = (1-x)(2-x)(3-x)\dots(n-x)$ at $x = 1$ is
1) $n!$ 2) $(n-1)!$
3) $-(n-1)!$ 4) 0
87. If $y = \sqrt{\frac{1+\cos 2\theta}{1-\cos 2\theta}}$ then $\frac{dy}{d\theta}$ at $\theta = \frac{3\pi}{4}$ is
1) -2 2) 2 3) 1 4) -1
88. If $\sqrt{1-x^{2n}} + \sqrt{1-y^{2n}} = a(x^n - y^n)$, then $\frac{\sqrt{1-x^{2n}} \frac{dy}{dx}}{\sqrt{1-y^{2n}} \frac{dy}{dx}}$ is equal to
1) $\frac{x^{n-1}}{y^{n-1}}$ 2) $\frac{y^{n-1}}{x^{n-1}}$ 3) $\frac{x}{y}$ 4) 1
89. If $y = |\cos x| + |\sin x|$, then $\frac{dy}{dx}$ at $x = \frac{2\pi}{3}$ is
1) $\frac{1-\sqrt{3}}{2}$ 2) 0
3) $\frac{\sqrt{3}-1}{2}$ 4) $\frac{\sqrt{3}+1}{2}$
90. If $f^1(x) = \sqrt{2x^2-1}$ and $y = f(x^2)$, then $\frac{dy}{dx}$ at $x = 1$ is
1) 2 2) 1 3) -2 4) -1
91. The first derivative of the function $\cos^{-1}\left(\sin\sqrt{\frac{1+x}{2}}\right) + x^x$ with respect to x at $x = 1$ is
1) $\frac{3}{4}$ 2) 0 3) $\frac{1}{2}$ 4) $-\frac{1}{2}$
92. If $f(\theta) = \sin\left(\tan^{-1}\left(\frac{\sin \theta}{\sqrt{\cos 2\theta}}\right)\right)$, where $-\frac{\pi}{4} < \theta < \frac{\pi}{4}$, then the value of $\frac{d}{d(\tan \theta)} f(\theta)$ is
1) 1 2) -1 3) 0 4) $\sqrt{2}$
93. If $y = \cos^{-1}(\cos x)$, then $\frac{dy}{dx}$ at $x = \frac{5\pi}{4}$ is
1) 1 2) -1 3) $\frac{1}{\sqrt{2}}$ 4) $\frac{5\pi}{4}$

94. If $y = f\left(\frac{3x+4}{5x+6}\right)$ and $f'(x) = \tan x^2$ then $\frac{dy}{dx}$ is equal to

- 1) $-2 \tan\left(\frac{3x+4}{5x+6}\right) \times \frac{1}{(5x+6)^2}$
 2) $f\left(\frac{3 \tan x^2 + 3}{5 \tan x^2 + 6}\right) \tan x^2$
 3) $\tan x^2$
 4) $2 \tan\left(\frac{3x+4}{5x+6}\right) \times \frac{1}{(5x+6)^2}$

95. If $x = \operatorname{cosec} \theta - \sin \theta$, $y = \operatorname{cosec}^n \theta - \sin^n \theta$ then $(x^2 + 4) \left(\frac{dy}{dx}\right)^2 - n^2 y^2 =$

- 1) n^2 2) $2n^2$ 3) $3n^2$ 4) $4n^2$

96. Differential coefficient of

$$\left(x^{\frac{l+m}{m-n}}\right)^{\frac{1}{n-l}} \cdot \left(x^{\frac{m+n}{n-l}}\right)^{\frac{1}{l-m}} \cdot \left(x^{\frac{n+l}{l-m}}\right)^{\frac{1}{m-n}} \text{ w.r.t. } x, \text{ is}$$

- 1) 1 2) 0 3) -1 4) x^{lmn}

97. Let $u(x)$ and $v(x)$ are differentiable function such that $\frac{u(x)}{v(x)} = 7$. If $\frac{u'(x)}{v'(x)} = p$

$$\text{and } \left(\frac{u(x)}{v(x)}\right)' = q, \text{ then } \frac{p+q}{p-q} =$$

- 1) 1 2) 0 3) 7 4) -7

98. If $f(x) = |\cos x - \sin x|$, then $f'\left(\frac{\pi}{4}\right) =$

- 1) $\sqrt{2}$ 2) $-\sqrt{2}$ 3) 0 4) does not exist

99. A function is represented parametrically by the equations $x = \frac{1+t}{t^3}$, $y = \frac{3}{2t^2} + \frac{2}{t}$ then

$$\frac{dy}{dx} - x \left(\frac{dy}{dx}\right)^2 \text{ has the absolute value equal to}$$

- 1) -1 2) 1 3) 0 4) 2

100. If $f(x) = \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix}$, then $f'(x) =$

- 1) 0 2) $6x$ 3) $12x^3$ 4) $6x^2$

101. Let y be an implicit function of x defined by $x^{2x} - 2x^x \cot y - 1 = 0$. Then $y'(1)$ equals [AIEEE-2009]

- 1) -1 2) 1 3) $\log 2$ 4) $-\log 2$

102. Let $f: (-1, 1) \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = -1$ and $f'(0) = 1$. Let $g(x) = [f(2f(x) + 2)]^2$, then $g'(0) =$

[AIEEE 2010]

- 1) -4 2) 0 3) -2 4) 4

103. Let f be twice differentiable function such that $g^1(x) = -f(x)$ and $f^1(x) = g(x)$, $h(x) = (f(x))^2 + (g(x))^2$. If $h(5) = 11$, then $h(10)$ is

- 1) 22 2) 11 3) 0 4) 8

104. Given that $f(x)$ is a differentiable function of x and that $f(x) \cdot f(y) = f(x) + f(y) + f(xy) - 2$ and that $f(2) = 5$. Then $f(3)$ is equal to

- 1) 10 2) 24 3) 15 4) 19

105. Let $f(x+y) = f(x)f(y)$ and $f(x) = 1 + (\sin 2x)g(x)$ where $g(x)$ is continuous. Then $f'(x)$ equals

- 1) $f(x)g(0)$ 2) $2f(x)g(0)$
 3) $2g(0)$ 4) $2f(0)$

LEVEL-III-KEY

- 1) 2 2) 3 3) 3 4) 4 5) 1
 6) 2 7) 2 8) 1 9) 3 10) 3
 11) 1 12) 4 13) 3 14) 2 15) 2
 16) 3 17) 1 18) 2 19) 4 20) 3
 21) 2 22) 1 23) 2 24) 3 25) 4
 26) 4 27) 1 28) 1 29) 3 30) 1
 31) 2 32) 3 33) 4 34) 1 35) 1
 36) 2 37) 3 38) 3 39) 2 40) 1
 41) 2 42) 3 43) 2 44) 2 45) 4
 46) 4 47) 4 48) 4 49) 1 50) 1
 51) 3 52) 1 53) 3 54) 1 55) 2
 56) 2 57) 1 58) 3 59) 1 60) 2

- 61) 1 62) 1 63) 4 64) 2 65) 3
 66) 2 67) 3 68) 3 69) 2 70) 4
 71) 3 72) 2 73) 3 74) 1 75) 4
 76) 2 77) 1 78) 2 79) 4 80) 3
 81) 1 82) 2 83) 3 84) 3 85) 2
 86) 3 87) 2 88) 1 89) 3 90) 1
 91) 3 92) 1 93) 2 94) 1 95) 4
 96) 2 97) 1 98) 4 99) 2 100) 4
 101) 102) 1 103) 2 104) 1 105) 2

LEVEL-III-HINTS

- $g(y) \log f(x) = f(x) - g(y)$

$$\Rightarrow \frac{dy}{dx} = \frac{f'(x) \cdot \log f(x)}{g'(y)(1 + \log f(x))^2}$$
- $\left(\frac{x^2 - y^2}{x^2 + y^2} \right) = \cos(\log a)$
 Applying Componendo and dividendo
- $f(x) = (x - a)g(x), f(a) = 0$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow a} \frac{(x - a)g(x) - 0}{x - a} = g(a)$$
- $\lim_{x \rightarrow y} \frac{|f(x) - f(y)|}{|x - y|} \leq |x - y|^2$
 $f'(x) = 0 \Rightarrow f(x)$ is constant
- Differentiate $\phi(x) = f(x)g(x)$ two times and simplify
- Since $f(x)$ is an even function $f'(0) = 0$
- $\frac{1}{x} = \frac{\sqrt{1+t^2}}{\sqrt{1-t^2}} \Rightarrow \frac{1-x}{1+x} = \frac{\sqrt{1+t^2} - \sqrt{1-t^2}}{\sqrt{1+t^2} + \sqrt{1-t^2}}$

$$\Rightarrow \frac{1-x}{1+x} = y$$
- $\frac{d}{dx} \{ \sec hx + \cosh x \}$

$$= -\sec hx \tanh x - \cosh x \coth x$$

$$= -\left[\frac{\sinh x}{\cosh^2 x} + \frac{\cosh x}{\sinh^2 x} \right] = -\left[\frac{\sinh^3 x + \cosh^3 x}{\cosh^2 x \sinh^2 x} \right]$$
- $(1+x)(1+x^2)\dots(1+x^n) = \frac{1-x^{2n}}{1-x}$ and
 taking logarithms and differentiate

- $f(x) = (1 + \tan x) \left(1 + \frac{1 - \tan x}{1 + \tan x} \right) = 2$
- Taking Logarithm on both sides, then

$$\frac{dy}{dx} = \frac{-\left(\frac{\partial f}{\partial x} \right)}{\left(\frac{\partial f}{\partial y} \right)}$$
- $\frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta} \right)}{\left(\frac{dx}{d\theta} \right)}$
- $P(0) = 4 \Rightarrow d = 4, P^1(x) = 3ax^2 + 2bx + c$
 at $x = 0, P^1(0) = 3$ then $c = 3$
 similarly to find a and b calculate $P^{11}(0)$ and $P^{111}(0)$
- Verify the options
- If $x > 4$ then $|x - 2| = x - 2$
- $\frac{d}{dx}(x^2 - x + 1) = ax + b, 2x - 1 = ax + b$
 $\Rightarrow a = 2, b = -1$
- Express 'y' as $\log \cos \tan^{-1}(\sinh x) = \log(\sec hx)$
 and simplify
- Express $\log \left\{ (e^x + 2) + \sqrt{(e^x + 2)^2 + 1} \right\}$ as
 $\sinh^{-1}(e^x + 2)$ and simplify
- $f(x) = (ax + b)\cos x + (cx + d)\sin x$

$$f'(x) = (ax + b)(-\sin x) + \cos x \cdot a + (cx + d)\cos x + c\sin x$$

 Compare on Both sides
- Find its determinant and put $x = \frac{\pi}{2}$.
- $\log(a^m) = m \log a,$

$$\frac{d}{dx}(\log x) = \frac{1}{x}, \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$
- $\frac{d}{dx}(f^2(x) + g^2(x)) = 2[f(x)f'(x) + g(x)g'(x)]$

$$= 2[f(x)g(x) - g(x)f(x)] = 0$$

 hence $f^2(x) + g^2(x)$ is constant. Thus

$$f^2(24) + g^2(24) = f^2(2) + g^2(2) = 16 + 16 = 32$$

23. $f(x) = \cos x \cdot \cos 3x \dots \cos(2n-1)x$
 $= \cos[x + 3x + \dots + (2n-1)x] = \cos[(1+3+\dots+(2n-1))x]$
 $= \cos(n^2x) \Rightarrow f(x) = \cos n^2x + i \sin n^2x$
differentiate w.r.t 'x' on both sides

24. consider $(\sqrt{x+y})^2 - (\sqrt{y-x})^2$
 $= (\sqrt{x+y} - \sqrt{y-x})(\sqrt{x+y} + \sqrt{y-x})$

$$2x = (\sqrt{x+y} - \sqrt{y-x}) \cdot c$$

$$\Rightarrow \sqrt{x+y} - \sqrt{y-x} = \frac{2x}{c} \rightarrow (1)$$

given $\sqrt{x+y} + \sqrt{y-x} = c \rightarrow (2)$

$$(1) + (2): 2\sqrt{x+y} = \frac{2x}{c} + c$$

$$4(x+y) = \frac{4x^2}{c^2} + c^2 + 4x, \text{ then } 4y = \frac{4x^2}{c^2} + c^2$$

differentiate w.r.t 'x' two times

25. $\frac{dy}{dx} = 2e^{2x}$ and $\frac{d^2y}{dx^2} = 4e^{2x}$. Also $x = \frac{1}{2} \log y$.

$$\text{so } \frac{dx}{dy} = \frac{1}{2y} \text{ and } \frac{d^2x}{dy^2} = -\frac{1}{2y^2} = -\frac{1}{2} e^{-4x}$$

$$\text{Hence } \left(\frac{d^2y}{dx^2} \right) \left(\frac{d^2x}{dy^2} \right) = -2e^{-2x}$$

26. $f(x) = \sqrt{x^2 - 2x + 1} = |x-1| = \begin{cases} x-1 & \text{if } x > 1 \\ 0 & \text{if } x = 1 \\ -(x-1) & \text{if } x < 1 \end{cases}$

$$f'(x) = 1 \text{ if } x > 1 \text{ and } f'(x) = -1 \text{ if } x < 1$$

27. Differentiating, we have,

$$\cos x \cos y - \sin x \sin y \frac{dy}{dx} = 0.$$

Putting $x = y = \frac{\pi}{4}$, we have $\frac{dy}{dx} \Big|_{(\frac{\pi}{4}, \frac{\pi}{4})} = 1$.

differentiating again, we have

$$-\sin x \cos y - \cos x \sin y \frac{dy}{dx} - \cos x \sin y \frac{dy}{dx}$$

$$-\sin x \sin y \left(\frac{dy}{dx} \right)^2 - \sin x \sin y \frac{d^2y}{dx^2} = 0,$$

Putting $x = y = \frac{\pi}{4}$, we have $\frac{d^2y}{dx^2} \Big|_{(\frac{\pi}{4}, \frac{\pi}{4})} = -4$

28. $f(x) = 0, x = \frac{\pi}{2}$

$$= 1, x < \frac{\pi}{2}$$

$$= -1, x > \frac{\pi}{2}$$

29. $y^2 = p(x) \Rightarrow 2yy_1 = p'(x)$ then $2yy_2 + 2y_1y_1 = p''(x)$
 $\Rightarrow 2[y^3y_2 + y^2y_1^2] = y^2p''(x)$

$$\Rightarrow 2 \left[y^3y_2 + \frac{(p'(x))^2}{4} \right] = y^2p''(x)$$

$$\Rightarrow 2y^3y_2 = y^2p''(x) - \frac{(p'(x))^2}{2}$$

$$\Rightarrow y^3y_2 = \frac{1}{2} \left[y^2p''(x) - \frac{(p'(x))^2}{2} \right]$$

differentiate w.r.t 'x' on both sides

30. $\cos^{-1} \frac{9-x^2}{9+x^2} = \cos^{-1} \frac{1-(\frac{x}{3})^2}{1+(\frac{x}{3})^2} = 2 \tan^{-1}(\frac{x}{3})$

If $0 \leq x < \infty$ and is equal to $-2 \tan^{-1}(\frac{x}{3})$

If $-\infty < x \leq 0$. Hence

$$y'(x) = \begin{cases} \frac{2}{1+(\frac{x}{3})^2} \cdot \frac{1}{3} & \text{if } 0 < x < \infty \\ -\frac{2}{1+(\frac{x}{3})^2} \cdot \frac{1}{3} & \text{if } -\infty < x < 0 \end{cases}$$

$$\text{Hence } y'(-1) = -\frac{3}{5}$$

31. $x = 2t - |t| = \begin{cases} t, & t \geq 0 \\ 3t, & t < 0 \end{cases}$ so $t = \begin{cases} x, & x \geq 0 \\ \frac{x}{3}, & x < 0 \end{cases}$

therefore, we can express $y = t^2 + t|t| = \begin{cases} 2t^2, & x \geq 0 \\ 0, & x < 0 \end{cases}$

$$= \begin{cases} 2x^2, & x \geq 0 \\ 0, & x < 0 \end{cases} \text{ Hence } y'(x) = \begin{cases} 4x, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Note that we cannot find $\frac{dy}{dx}$ as the derivative does not exist at $x=0$

32. Given $t = \sqrt{6} - \sqrt{4} - \sqrt{3} + \sqrt{2} = \tan 7\frac{1}{2}$

$$\therefore f(x) = \frac{2x}{1-x^2} \Rightarrow f(t) = \tan 15^\circ = (2 - \sqrt{3})$$

$$g(x) = \frac{3x - x^3}{1-3x^2} \Rightarrow g(t) = \tan\left(22\frac{1}{2}\right) = \sqrt{2} - 1$$

$$\text{Now } f(g(t)) = f(\sqrt{2} - 1) = \tan 2\left(22\frac{1}{2}\right) = \tan 45 = 1$$

$$\Rightarrow (f(g(t)))^1 = 0$$

33. $f(x) = \cos x \cos 2x \cos 4x \cos 8x \cos 16x$

$$= \cos A \cos 2A \cos 4A \dots \cos 2^n A = \frac{\sin 2^{n+1} A}{2^{n+1} \sin A}$$

$$= \frac{\sin 2^5 x}{2^5 \sin x} = \frac{\sin 32x}{32 \sin x}$$

$$f'(x) = \frac{32 \sin x \cos 32x - \cos x \sin 32x}{32 \sin^2 x}$$

$$\text{then } f'(\pi/4) = \frac{32 \sin \pi/4 \cos 8\pi - 0}{32(\pi/2)} = 2 \sin \pi/4 = \sec \pi/4$$

34. $f(x) = |x-8| = x-8$ as $x > 8$

$$\text{Now } g(x) = f(f(x)) = f(x-8) = f(X) = X-8$$

$$g(x) = f(f(x)) = x-16 \quad \therefore g'(x) = 1$$

35. Let $y^{y^v} = \log_e (x + \log_e (x + \dots)) = v$

$$\therefore y^v = \log_e (x + v) = v \quad \Rightarrow y = v^{1/v} \text{ and } x = e^v - v$$

$$\Rightarrow \frac{dy}{dv} = v^{1/v} \left(\frac{1}{v} - \frac{1}{v^2} \log v \right) \text{ and } \frac{dx}{dv} = (e^v - 1)$$

$$\text{i.e. } \frac{dy}{dv} = v^{1/v} \left(\frac{1}{v} - \frac{1}{v^2} \log v \right) = v^{1/v-2} (1 - \log v)$$

$$\text{and } \frac{dx}{dv} = (e^v - 1) \therefore \frac{dy}{dx} = \frac{v^{1/v-2} (1 - \log v)}{e^v - 1}$$

$$\text{at } x = e^2 - 2, y = \sqrt{2}, v^{1/v} = \sqrt{2}, e^v - v = e^2 - 2$$

$$\Rightarrow v = 2 \text{ or } 4 \text{ which is not valid}$$

$$\therefore v = 2, e^v - v = e^2 - 2 = x \text{ given so true}$$

$$\left(\frac{dy}{dx} \right)_{e^2-2, \sqrt{2}} = \left(\frac{dy}{dv} \right)_{v=2} = \frac{1 - \log 2}{2\sqrt{2}(e^2 - 1)}$$

36. $y = \frac{\sin^3 x}{\sin x + \cos x} + \frac{\cos^3 x}{\cos x + \sin x} = \frac{\sin^3 x + \cos^3 x}{\sin x + \cos x}$

$$= \sin^2 x + \cos^2 x - \sin x \cos x = 1 - \frac{1}{2} \sin 2x \text{ then}$$

$$y'(x) = -\cos 2x$$

37. By verification

38. $t = \sin^{-1}(2^s) \Rightarrow \sin t = 2^s$ differentiate w.r.t. 't' on

$$\text{both sides } \cos t = 2^s \log 2 \cdot \frac{ds}{dt}$$

$$\Rightarrow \cos t = \sin t \log 2 \cdot \frac{ds}{dt} \Rightarrow \frac{ds}{dt} = \frac{\cot t}{\log 2}$$

39. Express 'y' as $\frac{x}{x + \frac{x}{b+y}}$ and then differentiate

40. In the neighbourhood of $-\pi/4$, we have

$$f(x) = (-x)^{-\sin x} = e^{-\sin x \log(-x)}$$

$$\Rightarrow f'(x) = e^{-\sin x \log(-x)} \left(-\cos x \log(-x) - \frac{\sin x}{x} \right)$$

$$\Rightarrow f'(x) = (-x)^{-\sin x} \left(-\cos x \log(-x) - \frac{\sin x}{x} \right)$$

$$\Rightarrow f'(-\pi/4) = (\pi/4)^{1/\sqrt{2}} \left(\frac{-1}{\sqrt{2}} \log(\pi/4) + \frac{4}{\pi} \times \frac{-1}{\sqrt{2}} \right)$$

$$= (\pi/4)^{1/\sqrt{2}} \left(\frac{\sqrt{2}}{2} \log(\pi/4) - \frac{2\sqrt{2}}{\pi} \right)$$

41. $\tan^{-1}(x+1) - \tan^{-1}x + \tan^{-1}(x+2)$

$$- \tan^{-1}x + 1 - \dots = \tan^{-1}x + n - \tan^{-1}x$$

42. Put $x = \sin \alpha, y = \sin \beta$

43. we have $f(x) = 3e^{x^2}$, differentiating w.r.t. x, we get $f'(x) = 6xe^{x^2}$

$$\text{Now } f'(x) = 6xe^{x^2} - 2xf'(x) + \frac{1}{3}f(0) - f'(0)$$

$$= 6xe^{x^2} - 6xe^{x^2} + \frac{1}{3}(3) - 0 = 1$$

$$\Rightarrow f(0) = 3 \text{ and } \Rightarrow f'(0) = 0$$

44. $\frac{dy}{dx} = f'\left(\frac{2x+3}{3-2x}\right) \cdot \frac{d}{dx}\left(\frac{2x+3}{3-2x}\right)$

$$= \sin\left[\log\left(\frac{2x+3}{3-2x}\right)\right] \cdot \frac{12}{(3-2x)^2}$$

45. $f(x) = \frac{\log(\log x)}{\log x^2}$ then differentiate w.r.t 'x' on both sides and substitute $x = e$

46. $n=2$, verification

47. $\frac{d}{dy}\left(\frac{dx}{dy}\right) = \frac{d}{dy}\left(\frac{1}{\left(\frac{dy}{dx}\right)}\right)$

48. $\frac{dy}{dx} = n \left(x + \sqrt{x^2 - a^2} \right)^{n-1} \cdot \left(1 + \frac{1}{2\sqrt{x^2 - a^2}} \cdot 2x \right)$

$$= n \frac{\left(x + \sqrt{x^2 - a^2} \right)^n}{\left(x + \sqrt{x^2 - a^2} \right)} \left(\frac{x + \sqrt{x^2 - a^2}}{\sqrt{x^2 - a^2}} \right)$$

49. $(1-x)^{-p/q} = 1 + p \left(\frac{x}{q} \right) + \frac{p(p+q)}{2!} \left(\frac{x}{q} \right)^2 + \dots$

$$\Rightarrow \frac{d}{dx} (1-x)^{-p/q}$$

50. $\frac{d}{dx} \left(\frac{\tan x - 1/\tan x}{\tan x + 1/\tan x} \right) = \frac{d}{dx} \left(\frac{-(1 - \tan^2 x)}{1 + \tan^2 x} \right)$

$$= \frac{d}{dx}(-\cos 2x)$$

$$51. \quad y = \tan^{-1}\left(\frac{1}{1+x(x-1)}\right) = \tan^{-1}\left(\frac{x-(x-1)}{1+x(x-1)}\right)$$

$$52. \quad \frac{d}{dx}(\sec^{-1}x) = \frac{1}{x\sqrt{x^2-1}}$$

$$53. \quad \frac{d}{dx}\left(\coth\left(\frac{x}{2}\right)\right) = -\operatorname{cosech}^2\left(\frac{x}{2}\right) \cdot \left(\frac{1}{2}\right)Z$$

$$54. \quad \frac{dy}{d\theta} = n \sec^{n-1}\theta \cdot \sec\theta \tan\theta - n \cos^{n-1}\theta(-\sin\theta) \\ = n \tan\theta (\sec^n\theta + \cos^n\theta)$$

$$\frac{dx}{d\theta} = \sec\theta \tan\theta + \sin\theta$$

$$\left(\frac{dy}{dx}\right)^2 = \left[\frac{n \tan\theta (\sec^n\theta + \cos^n\theta)}{\tan\theta (\sec\theta + \cos\theta)}\right]^2$$

$$= \frac{n^2 (\sec^n\theta + \cos^n\theta)^2}{(\sec\theta + \cos\theta)^2}$$

$$= \frac{\left[n^2 (\sec^n\theta - \cos^n\theta)^2 + 4 \sec^n\theta \cos^n\theta\right]}{(\sec\theta - \cos\theta)^2 + 4 \sec\theta \cos\theta}$$

$$= \frac{n^2(y^2+4)}{x^2+4}$$

$$55. \quad \text{Express 'y' as } \cot x - \cot(n+1)x \text{ and differentiate}$$

$$56. \quad 3f(x) - 2f\left(\frac{1}{x}\right) = x \quad \text{-----(1)}$$

$$\text{Put } X = \frac{1}{x} \text{ we get}$$

$$3f\left(\frac{1}{x}\right) - 2f(x) = \frac{1}{x} \quad \text{----- (2)}$$

Solve (1) and (2).

$$57. \quad \log x = \tan^{-1}\left(\frac{y-x^2}{x^2}\right)$$

$$y = x^2 + x^2 \tan(\log x)$$

$$58. \quad \text{Eliminate } f(\tan x)$$

$$f(\cot x) = \frac{x}{b-a} - \frac{a\pi}{2(b^2-a^2)}$$

$$59. \quad \frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx}$$

$$60. \quad \text{Let } g(x) \text{ be the inverse of } f(x)$$

$$\Rightarrow g^1(f(x))f^1(x) = 1$$

$$g^1(f(-1))f^1(-1) = 1 \Rightarrow g^1(-2) = \frac{1}{14}$$

$$61. \quad f(x) = \cos x \cos 2x \cos 2^2 x \cos 2^3 x \dots \cos 2^{n-1} x$$

$$\Rightarrow f(x) = \frac{\sin 2^n x}{2^n \sin x}$$

$$\Rightarrow f^1(x) = \frac{2^n \cos 2^n x \sin x - \sin 2^n x \cos x}{2^n \sin^2 x}$$

$$\Rightarrow f^1\left(\frac{\pi}{2}\right) = \frac{2^n \cos 2^{n-1} \pi}{2^n} = \cos 2^{n-1} \pi = (-1)^{2^{n-1}} = 1$$

$$62. \quad g(x) = f^{-1}(x) \Rightarrow f(g(x)) = x$$

$$\Rightarrow f^1(g(x)) \cdot g^1(x) = 1$$

$$63. \quad y = \{x^{11} + f(x)\}^{-2}$$

$$64. \quad (fog)(x) = x \text{ for all } x$$

$$\Rightarrow f^1(g(x))g^1(x) = 1 \text{ for all } x$$

$$\Rightarrow f^1(g(1))g^1(1) = 1$$

$$\Rightarrow g^1(1) = \frac{1}{f^1(g(1))}$$

$$\text{But } (gof)(x) = x \Rightarrow g(f(0)) = 0$$

$$\Rightarrow g(1) = 0 \text{ and } f^1(0) = \frac{1}{2}$$

$$\text{Hence, } g^1(1) = \frac{1}{\frac{1}{2}} = 2$$

$$65. \quad \frac{dy}{dx} = -\frac{f_x}{f_y}$$

$$66. \quad f'(x) = \begin{vmatrix} 0 & 0 & 0 \\ \theta \sec x & \tan x & x \\ 1 & \tan x - \tan \theta & \theta \end{vmatrix}$$

$$+ \begin{vmatrix} \sec \theta & \tan^2 \theta & 1 \\ \theta \sec x \tan x & \sec^2 x & 1 \\ 1 & \tan x - \tan \theta & 0 \end{vmatrix} + \begin{vmatrix} \sec \theta & \tan^2 \theta & 1 \\ \theta \sec \theta & \tan \theta & \theta \\ 0 & \sec^2 x & 0 \end{vmatrix}$$

$$f'(0) = 0 + \begin{vmatrix} \sec \theta & \tan^2 \theta & 1 \\ \theta \sec \theta \tan \theta & \sec^2 \theta & 1 \\ 1 & 0 & 0 \end{vmatrix} + \begin{vmatrix} \sec \theta & \tan^2 \theta & 1 \\ \theta \sec \theta & \tan \theta & \theta \\ \theta & \sec^2 \theta & 0 \end{vmatrix} = -1$$

$$67. \quad \frac{du}{dx} = f'(x^2)(2x) = (\sin x^2)(2x)$$

$$\frac{du}{dx} = g'(x^3)(3x^2) = \cos(x^3)(3x^2)$$

$$\frac{du}{dv} = \frac{du/dx}{dv/dx} = \frac{2x \sin x^2}{3x^2 \cos x^3} = \frac{2 \sin x^2}{3x \cos x^3}$$

$$68. \quad \frac{dy}{dx} = f'[f(f(f(x)))] \cdot f'[f(f(x))] \cdot f'[f(x)] \cdot f'(x)$$

$$\left(\frac{dy}{dx}\right)_{x=0} = f'[f(f(f(0)))] \cdot f'[f(f(0))] \cdot f'[f(0)] \cdot f'(0)$$

$$= f'[f(f(0))] \cdot f'[f(0)] \cdot f'(0) \cdot (2)$$

$$= f'[f(0)] \cdot f'(0) \cdot (2) \cdot (2)$$

$$= f'(0) \cdot (2) \cdot (2) \cdot (2) \cdot (2) = 16$$

$$69. \quad \frac{dy}{dx} = \frac{\left(\frac{dy}{dx}\right)}{\left(\frac{dx}{dt}\right)} = \frac{\psi^1}{\phi^1}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{\psi^1}{\phi^1} \right) = \frac{d}{dt} \left(\frac{\psi^1}{\phi^1} \right) \frac{dt}{dx} = \frac{\phi^1 \psi^{11} - \psi^1 \phi^{11}}{(\phi^1)^2} \cdot \frac{1}{\phi^1}$$

$$= \frac{\phi^1 \psi^{11} - \psi^1 \phi^{11}}{(\phi^1)^3}$$

$$70. \quad \frac{dx}{ds} = 1, \frac{dy}{ds} = 2, \frac{d^2x}{ds^2} = 0, \frac{d^2y}{ds^2} = 0$$

$$u = x^2 + y^2 \Rightarrow \frac{du}{ds} = 2x \frac{dx}{ds} + 2y \frac{dy}{ds}$$

$$\frac{d^2u}{ds^2} = \frac{d}{ds} \left[\frac{du}{ds} \right] = \frac{d}{ds} \left[2x \frac{dx}{ds} + 2y \frac{dy}{ds} \right]$$

$$= 2 \left(\frac{dx}{ds} \right)^2 + 2x \frac{d^2x}{ds^2} + 2 \left(\frac{dy}{ds} \right)^2 + 2y \left(\frac{d^2y}{ds^2} \right)$$

$$= 2(1) + 0 + 2(4) + 0 = 10$$

$$71. \quad t^8 = x - 1 \Rightarrow t^2 = (x - 1)^{1/4}$$

$$y = (t)^2 t^8 + 1 \Rightarrow y = (x - 1)^{5/4} + 1 \Rightarrow$$

$$\frac{dy}{dx} = \frac{5}{4} (x - 1)^{1/4}$$

$$\frac{d^2y}{dx^2} = \frac{5}{16} (x - 1)^{-3/4} =$$

$$\frac{5}{16(x - 1)^{3/4}} = \frac{5}{16(t^2)^3} = \frac{5}{16t^6}$$

$$72. \quad \frac{d}{dx} \left[\cos^2 \left(\tan^{-1} \left(\sin \sin^{-1} \frac{1}{\sqrt{1+x^2}} \right) \right) \right]$$

$$\frac{d}{dx} \left[\cos^2 \left[\tan^{-1} \frac{1}{\sqrt{1+x^2}} \right] \right] = \frac{d}{dx} \left[\frac{1 + \cos 2 \left(\tan^{-1} \frac{1}{\sqrt{1+x^2}} \right)}{2} \right]$$

$$= \frac{d}{dx} \left[1 + \frac{1 - \frac{1}{1+x^2}}{1 + \frac{1}{1+x^2}} \right] = \frac{2x}{(2+x^2)^2}$$

$$73. \quad x \sin y = \sin y \cos a + \cos y \sin a$$

$$\Rightarrow \sin y (x - \cos a) = \cos y \sin a$$

$$\frac{\sin y}{\cos y} = \frac{\sin a}{x - \cos a} \Rightarrow y = \tan^{-1} \left(\frac{\sin a}{x - \cos a} \right)$$

$$74. \quad f(x) = \tan^{-1}(\cos x + n) - \tan^{-1}(\cos x)$$

$$f'(x) = \frac{1}{1+(n+\cos x)^2} (-\sin x) - \frac{1}{1+\cos^2 x} (-\sin x)$$

$$f'(0) = 0$$

$$75. \quad f'(x) = \frac{-1}{1 + \left(\frac{x^x - x^{-x}}{2} \right)^2} \frac{d}{dx} \left(\frac{x^x - x^{-x}}{2} \right)$$

$$f'(1) = \frac{-4}{(1+1)^2} \cdot \frac{1(1+\log 1) + 1(1+\log 1)}{2} = (-1)$$

$$\left(\frac{2}{2} \right) = -1$$

$$76. \quad \log y = x [\log(1 + 1/x)]$$

$$\frac{1}{y} y_1 = \frac{x^2}{x+1} \left(\frac{-1}{x^2} \right) + \log \left(1 + \frac{1}{x} \right)$$

$$\Rightarrow y_1 = \frac{-y}{x+1} + y \log \left(1 + \frac{1}{x} \right)$$

$$y(2) = \frac{9}{4}; y_1(2) = \frac{9}{4} \left(\frac{-1}{3} + \log \frac{3}{2} \right)$$

$$y_2(2) = y_1(2) \left(\frac{-1}{3} + \log \frac{3}{2} \right) + y(2) \left(\frac{1}{9} - \frac{1}{6} \right)$$

$$y_2(2) = \frac{9}{4} \left(\frac{-1}{3} + \log \frac{3}{2} \right)^2 - \frac{1}{8}$$

$$77. \quad y = \frac{f(x)}{1 + \frac{g(x)}{1 + \frac{f(x)}{1 + \dots \infty}}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{(1+y)f'(x) - yg'(x)}{1+2y+g(x)-f(x)}$$

$$78. \quad v+u=6\sqrt{uv} \Rightarrow v^2+u^2=34uv \text{ then differentiate}$$

$$79. \quad y = \tan^{-1} \left(\frac{8 \sin x \cos x}{\cos^2 x - 7 \sin^2 x} \right)$$

$$= \tan^{-1} \left(\frac{8 \tan x}{1-7 \tan^2 x} \right) = \tan^{-1} \left(\frac{7 \tan x + \tan x}{1-7 \tan x \cdot \tan x} \right)$$

$$y = \tan^{-1} (7 \tan x) + x$$

$$80. \quad \text{LEVEL - IV} \quad y = \tan^{-1} \left(\frac{\log x^2}{1+\log x^2} \right) + \tan^{-1} \left(\frac{3+2 \log x}{1-3(2 \log x)} \right)$$

$$= \tan^{-1}(1) - \tan^{-1}(\log x^2) + \tan^{-1}(3) + \tan^{-1}(3 \log x)$$

$$= \tan^{-1}(1) + \tan^{-1}(3)$$

$$\frac{d^2 y}{dx^2} = 0$$

$$81. \quad y = \cot^{-1} \left[\frac{2+2 \cos x}{2 \sin x} \right] = \cot^{-1} \left[\frac{1+\cos x}{\sin x} \right] =$$

$$\cot^{-1} \left[\cot \frac{x}{2} \right] = \frac{x}{2}; \quad \frac{dy}{dx} = \frac{1}{2}$$

$$82. \quad y^1 = -\frac{x}{y} \Rightarrow y^{11} = \frac{-y+xy^1}{y^2} = -\frac{2}{y^3}$$

$$-\frac{2}{y^3} + \frac{A}{y^3} = 0 \Rightarrow A = 2$$

$$83. \quad p(x^2+1) = (p(x))^2 + 1 \Rightarrow p(x) = x,$$

$$p^1(x) = 1 \Rightarrow p^1(0) = 1$$

$$84. \quad e^{x^e} \cdot e \cdot x^{e-1} + x^{e^x} \left[e^x \cdot \frac{1}{x} + e^x \cdot \log x \right]$$

$$+ e^{x^x} [x^x (1 + \log x)]$$

$$85. \quad \text{Taking logarithms on both sides \& differentiating}$$

$$86. \quad \text{Taking logarithms on both sides \& differentiating}$$

$$87. \quad y = |\cot \theta| = -\cot \theta, \left(\theta = \frac{3\pi}{4} \right)$$

$$88. \quad \text{Put } x^n = \sin \alpha, y^n = \sin \beta$$

$$89. \quad y = -\cos x + \sin x$$

$$90. \quad y = f(x^2) \Rightarrow \frac{dy}{dx} = f'(x^2) \cdot 2x$$

$$= 2x \sqrt{2x^4 - 1}$$

$$91. \quad f(x) = \frac{\pi}{2} - \sqrt{\frac{1+x}{2}} + x^x$$

$$92. \quad \text{Differentiate}$$

$$93. \quad [\because 0 \leq \cos^{-1} x \leq \pi \Rightarrow \text{in the neighbourhood of}$$

$$x = \frac{5\pi}{4} \text{ we have } 0 < 2\pi - x < \pi]$$

$$94. \quad \frac{dy}{dx} = f' \left(\frac{3x+4}{5x+6} \right) \frac{d}{dx} \left(\frac{3x+4}{5x+6} \right)$$

$$95. \quad \frac{dy}{dx} = \frac{d\theta}{dx} \cdot \frac{d\theta}{d\theta}$$

$$96. \quad \text{Exponent of}$$

$$x = \frac{l^2 - m^2 + m^2 - n^2 + n^2 - l^2}{(l-m)(m-n)(n-l)} = 0$$

$$y = x^0 = 1$$

$$97. \quad u(x) = 7 \cdot v(x) \Rightarrow u^1(x) = 7v^1(x) \Rightarrow p = 7$$

$$\left(\frac{u(x)}{v(x)} \right)^1 = 0 \Rightarrow q = 0$$

98. $f(x) = \cos x - \sin x, 0 < x \leq \frac{\pi}{4}$

$$\Rightarrow \sin x - \cos x, \frac{\pi}{4} < x < \frac{\pi}{2}$$

99. $\frac{dx}{dt} = -\left(\frac{3}{t^4} + \frac{2}{t^3}\right) = -\left(\frac{3+2t}{t^4}\right)$

$$\frac{dy}{dt} = -\left(\frac{3}{t^3} + \frac{2}{t^2}\right) = -\left(\frac{3+2t}{t^3}\right)$$

$$\frac{dy}{dx} = t$$

100. $f^1(x) = 0 + 0 + \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 0 & 6 \end{vmatrix}$

101. Put $x = 1 \Rightarrow y = \frac{\pi}{2}$ and then differentiate.

102. $g^1(x) = 2(f(2f(x)+2))\left(\frac{d}{dx}(f(2f(x)+2))\right)$
 $= 2f(2f(x)+2)f^1(2f(x)+2) \cdot (2f^1(x))$
 $= g^1(0) = 2f(2f(0)+2) \cdot f^1(2f(0)+2) \cdot 2(f^1(0))$
 $= 4f(0)f^1(0)$
 $= 4(-1)(1) = -4$

103. $h^1(x) = 2f(x)f^1(x) + 2g(x)g^1(x)$

as $f^1(x) = g(x), g^1(x) = -f(x)$

$$\therefore h^1(x) = 0$$

$$\therefore h(x) \text{ is constant function.}$$

104. we have, $f(x)f(y) = f(x) + f(y) + f(xy) - 2$

replace y by $\frac{1}{x}$

$$\Rightarrow f(x)f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right) + f(1) - 2$$

$$\Rightarrow f(x)f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$$

(since $f(1) = 2$ putting $x=y=1$)

$$\Rightarrow f(x) = x^n + 1 \Rightarrow f(2) = x^2 + 1 \text{ (since } f(2)=5)$$

$$\Rightarrow n = 2 \quad \therefore f(x) = x^2 + 1 \Rightarrow f(3) = 10$$

105. $f^1(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x)f(h) - f(x)}{h}$

$$= f(x) \lim_{h \rightarrow 0} \frac{f(h) - 1}{h} = f(x) \lim_{h \rightarrow 0} \frac{1 + (\sin 2h)g(h) - 1}{h}$$

$$= f(x) \lim_{h \rightarrow 0} \frac{\sin 2h}{h} \cdot \lim_{h \rightarrow 0} g(h) = 2f(x)g(0)$$



1. A: $\frac{d}{dx} \left(\tan^{-1} x + \sin^{-1} \frac{x}{\sqrt{1+x^2}} \right) \text{ at } x = 0$

B: $\frac{d}{dx} \left(\sec^{-1} \left(\frac{x^2 + \sqrt{x} + 1}{x^2 - \sqrt{x} - 1} \right) + \sin^{-1} \left(\frac{x^2 - \sqrt{x} - 1}{x^2 + \sqrt{x} + 1} \right) \right)$

C: $\frac{d}{dx} \left[\tan^{-1} \left(\frac{1 + \sin x}{\cos x} \right) \right]$

D: $\frac{d}{dx} \left(\cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) \right) \text{ at } x = 1$

Arrangement of the above values in the increasing order of the magnitude

- 1) B, C, D, A 2) B, A, D, C
 3) C, D, A, B 4) D, A, B, C

2. **Observe the following lists.**

List - I

List - II

A. The derivative of

$$\tan^{-1} \left(\frac{\sqrt{1+x^2} - 1}{x} \right) \quad 1) \frac{1}{1+x^2}$$

w.r.t. $\tan^{-1} x$

B. $\frac{d}{dx} \left(x\sqrt{x^2-9} - 9\cos h^{-1} \left(\frac{x}{3} \right) \right) \quad 2) \frac{2}{x}$

C. $\frac{d}{dx} \left(\tan^{-1} \left(\frac{2+3x}{3-2x} \right) \right) \quad 3) 2\sqrt{x^2-9}$

D. The derivative of

$$\sec^{-1} \left(\frac{1}{2x^2-1} \right) \text{ w.r.t. } \sqrt{1-x^2} \text{ is } 4) \frac{1}{2}$$

5. $\sqrt{x^2-9}$

The correct match for list - I from list - II is

	A	B	C	D
1)	5	4	2	3
2)	4	5	3	2
3)	3	4	1	2
4)	4	3	1	2

3. Statement I : If $\frac{d}{dx} \left(\frac{1}{f(x)} \right) = -\frac{f'(x)}{f^2(x)}$

then $\frac{d}{dx} \left(\frac{1}{f(x)} \right) = -\frac{f'(x)}{f^2(x)}$

Statement II : If $f(x)$ is constant then $f'(x) = 0$

Which of the above statements is correct

- 1) Only I 2) Only II
3) Both I and II 4) Neither I nor II

4. Statement I : If $x = \sin \theta + \cos \theta$,

$y = \cos \theta - \sin \theta$, then $\frac{dy}{dx} =$

Statement II : If $x = \sec \theta - \cos \theta$,

then $\frac{dy}{dx} =$

- 1) Only I is true 2) Only II is true
3) Both I and II are true
4) neither I nor II true

5. Statement I :

If $\frac{d}{dx} \left(\frac{1}{f(x)} \right) = -\frac{f'(x)}{f^2(x)}$
then the value of $\frac{d}{dx} \left(\frac{1}{f(x)} \right) =$

Statement II : Differential coefficient of the function

$f(g(x))$ w.r.t to the function $g(x)$ is $f'(g(x)) \cdot g'(x)$

Which of the above statements is correct

- 1) I only 2) II only
3) Both I and II 4) Neither I nor II

6. Statement I : If $y = \sin x + \cos x$ then

Statement II : If $y = |\cos x| + |\sin x|$ then

the value of $\frac{dy}{dx}$ is

Which of the above statments is correct

- 1) I only 2) II only
3) Both I and II 4) Neither I nor II

7. Observe the following

List - I

List - II

A. If $\sin^{-1} x + \sin^{-1} y = \frac{\pi}{2}$ then $\frac{dy}{dx} =$ 1) $\frac{1}{x}$

B. $\tan^{-1} x + \tan^{-1} y = \frac{\pi}{4}$ then $\frac{dy}{dx} =$ 2) $\frac{1}{1+x^2}$

C. If $x^m y^n = (x+y)^{m+n}$ then $\frac{dy}{dx} =$ 3) $-\frac{1}{x}$

D. If $\sin(x+y) = \log(x+y)$ then $\frac{dy}{dx} =$ 4) $\frac{1}{1+\cos(x+y)}$

then

The correct match is

	A	B	C	D
1)	2	1	3	4
2)	2	1	4	3
3)	1	2	3	4
4)	2	1	3	5

8. Assertion (A) : If

$f(x) = \frac{1}{x}$, then $f'(x) = 0$

Reason (R) : Derivative of constant function is zero

- 1) Both A & R are true, R is correct explanation of A
2) Both a & R are true, R is not correct explanation for A
3) A is true but R is false 4) A is false but R is true

9. a) $y = 2 \sin^{-1} x$ p) $\frac{dy}{dx} = \frac{2}{\sqrt{1-x^2}}$

b) $y = 4 \tan^{-1} x$ q) $\frac{dy}{dx} = \frac{4}{1+x^2}$

c) $y = 4 \sin^{-1} x$ r) $\frac{dy}{dx} = \frac{4}{\sqrt{1-x^2}}$

d) $y = 2 \cos^{-1} x$ s) $\frac{dy}{dx} = -\frac{2}{\sqrt{1-x^2}}$

10. **Observe the following lists.**
- | a | b | c | d | a | b | c | d |
|------|---|---|---|------|---|---|---|
| 1) p | q | r | s | 2) q | r | p | s |
| 3) r | s | p | q | 4) s | r | q | p |
- List - I**
- A.
- B.
- C.
- D.
- List - II**
- 1)
- 2)
- 3)
- 4)
- 5)
- The correct match for List - I from List - II is

- | A | B | C | D | A | B | C | D |
|------|---|---|---|------|---|---|---|
| 1) 3 | 5 | 2 | 1 | 2) 4 | 3 | 5 | 1 |
| 3) 3 | 5 | 1 | 2 | 4) 4 | 2 | 5 | 3 |
11. **Observe the following lists.**
- List - I**
- A. If $x = a(1 - \cos t)$; $y = a(t + \sin t)$ then at
- B.
- C.
- D.
- List - II**
- 1)
- 2)
- 3)
- 4) 1
- 5)
- The correct match for List - I from List - II is

- | | A | B | C | D |
|-----|---|---|---|---|
| 1) | 4 | 1 | 3 | 5 |
| 2) | 3 | 1 | 2 | 4 |
| 3). | 4 | 1 | 2 | 3 |
| 4) | 4 | 1 | 2 | 5 |

12. A:
- B:
- C: For the curve at
- is
- D:
- Arrangement of the above values in the increasing order of the magnitude
- 1) A, B, C, D 2) C, A, D, B
- 3) C, D, B, A 4) A, D, B, C
13. A: If $x = ct$, , at $t = 1$ is
- B: $x = 3 \cos$ - , $y = 3$
- Then is
- C: If then
-
- D: Derivative of $\log(\sec x)$ with respect to $\tan x$ at is
- Arrangement of the above values in the increasing order of the magnitude
- 1) C, D, A, B 2) C, A, D, B
- 3) A, B, D, C 4) B, D, A, C
14. A:
- B:
- C: at $x = 0$
- D:

Arrangement of the above values in the increasing order of the magnitude

- 1) B, C, A, D 2) D, A, B, C
3) D, B, C, A 4) A, B, C, D

LEVEL-IV - KEY

- 1) 1 2) 4 3) 2 4) 2 5) 3
6) 2 7) 2 8) 1 9) 2 10) 1
11) 3 12) 2 13) 3 14) 2

LEVEL-IV - HINTS

- A) Put $x = \tan \theta$

B) $\frac{d}{dx}(\sin^{-1}x + \cos^{-1}x) = \frac{d}{dx}\left(\frac{\pi}{2}\right)$

C) $\frac{d}{dx}\{\tan^{-1}(\sec x + \tan x)\} = \frac{1}{2}$

D) Put $x = \tan \theta$
- A) Put $x = \tan \theta$

B) $\frac{d}{dx}(\cos h^{-1}x) = \frac{-1}{\sqrt{x^2 - 1}}$

C) $\tan^{-1}\left(\frac{2}{3}\right) + \tan^{-1}(x)$

D) Put $x = \cos \theta$
- Differentiate and put $x = e$
- St - I & St - II : $\frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)}$
- St - I : taking logarithms on both sides and then differentiate it
St - II : By the formula (42), it is correct.
- St - I : By the formula (33)
St - II : Differentiate w.r.t. x
- A) $\sin^{-1}x = \frac{\pi}{2} - \sin^{-1}y$

B)

C) By taking logarithms and differentiate it

D)

$$8. \quad A : f(x) = \cos^2 x + \cos^2\left(x + \frac{\pi}{3}\right) - \cos x \cdot \cos\left(x + \frac{\pi}{3}\right)$$

$$= 1 - \cos\left(2x + \frac{\pi}{3}\right)$$

$$= \cos \frac{\pi}{3} - \frac{1}{2} \left[\cos\left(2x + \frac{\pi}{3}\right) + \cos \frac{\pi}{3} \right]$$

$$= 1 + \frac{1}{2} \cos\left(2x + \frac{\pi}{3}\right) - \frac{1}{2} \cos\left(2x + \frac{\pi}{3}\right)$$

$$-\frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4} \text{ then } f'(x) = 0$$

R : Derivative of constant function is zero

- Apply inverse trigonometric formulae
- A) Put $x = \sin \theta$ B) formula
C) Put $x = \sin \theta$ D) Put $x = \tan \theta$

$$11. \quad A) \frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)}$$

$$B) \frac{d}{dx}(x^x) = x^x (1 + \log x)$$

$$C) \text{ Put } x = a \sin \theta \quad D) \text{ Put } \sqrt{x} = \tan \theta$$

$$12. \quad A) \text{ put } x = \cos \theta, B) \frac{d}{dx} \left\{ \tan\left(\frac{\pi}{4} + x\right) \right\}$$

$$C) \frac{dy}{dx} = - \frac{\left(\frac{\partial f}{\partial x}\right)}{\left(\frac{\partial f}{\partial y}\right)}$$

D)

13.

A)



B)



$$C) \frac{dy}{dx} = \frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)} \quad D) \frac{dy}{dz} = \frac{\left(\frac{dy}{dx}\right)}{\left(\frac{dz}{dx}\right)}$$

14.

By the Formulae : A) formula 5, B) formula 13,

C) By the expansion of e^x and find its derivative by the formula (2)

$$D) \frac{d}{dx}(x^x) = x^x (1 + \log x)$$
